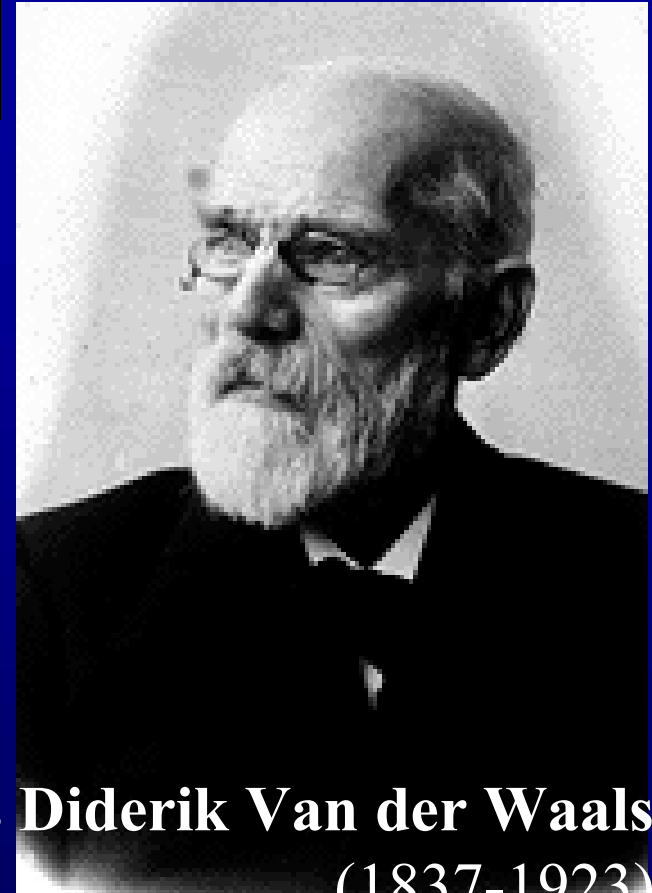


Chapter 14

Phase Transition

14.1 Van der Waals' equation (1873)

$$\left[p + a \left(\frac{n}{V} \right)^2 \right] (V - nb) = nRT$$



Johannes Diderik Van der Waals
(1837-1923)

The Nobel Prize in Physics 1910

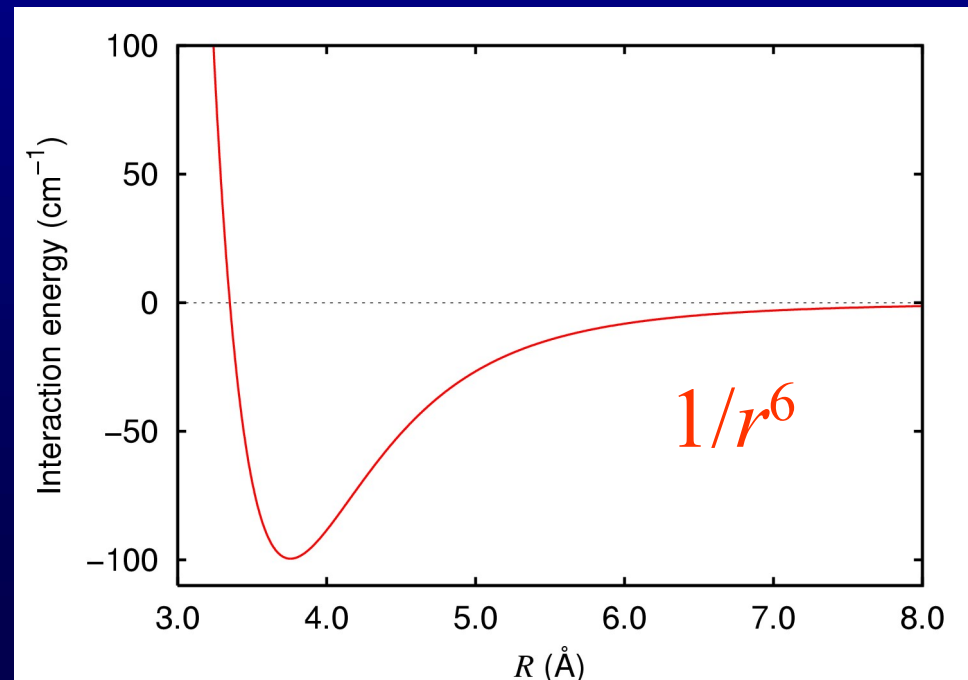
**"for his work on the equation of state for
gases and liquids"**

$$\left[p + a \left(\frac{n}{V} \right)^2 \right] (V - nb) = nRT$$

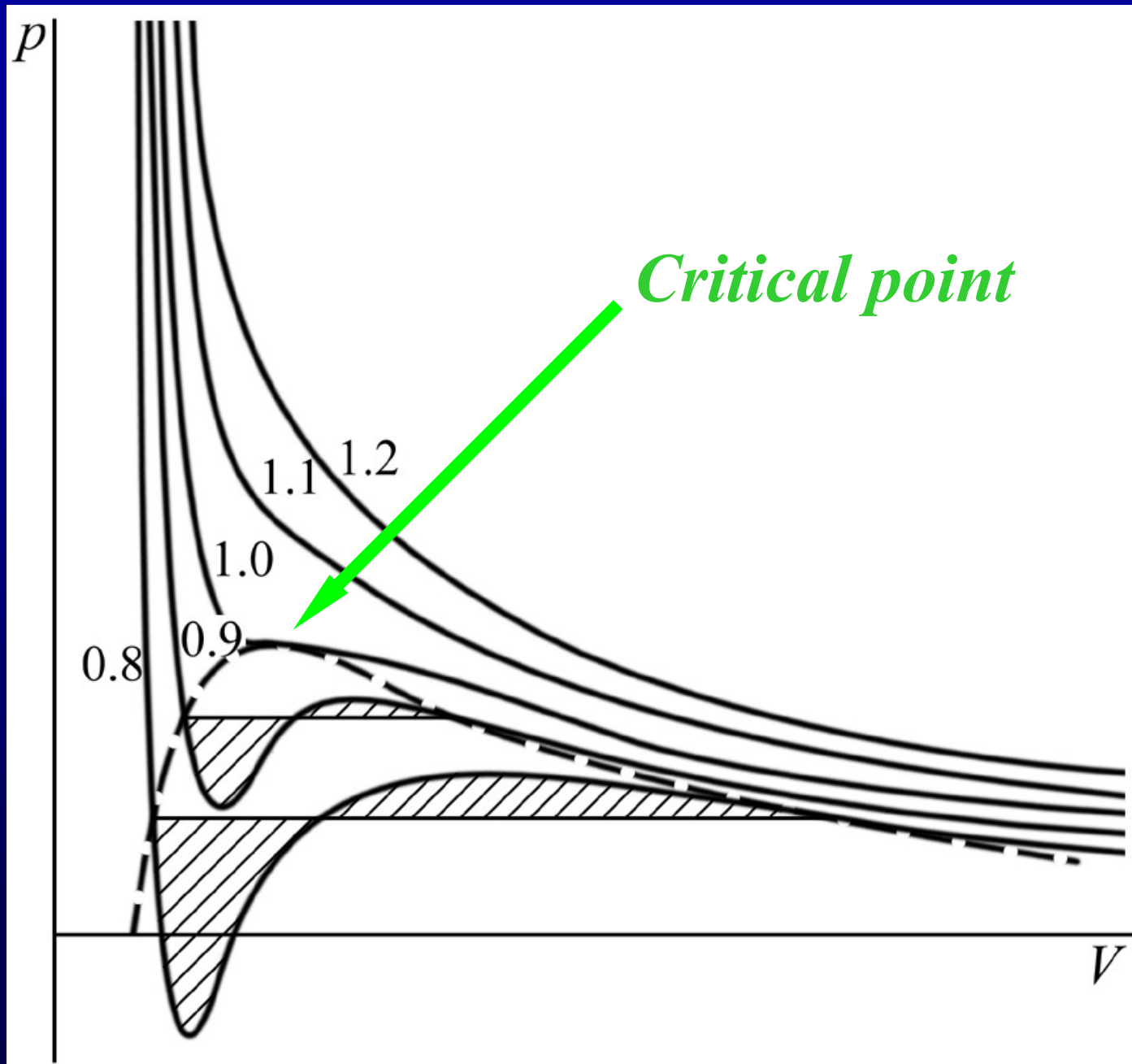
$$p = \frac{nRT}{V}$$

$$\frac{n}{V} \leftrightarrow \frac{n}{V}$$

density ----- density



Interaction energy of argon dimer.



$$\left(\frac{\partial p}{\partial V}\right)_T = 0, \quad \left(\frac{\partial^2 p}{\partial V^2}\right)_T = 0 \quad p_C, V_C, T_C$$

$$p = \frac{nRT}{V - nb} - a \left(\frac{n}{V}\right)^2 \quad \dots$$

Dimensionless version:

$$\left(\tilde{p} + \frac{3}{\tilde{V}^2}\right)(3\tilde{V} - 1) = 8\tilde{T}$$

tilde

where

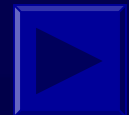
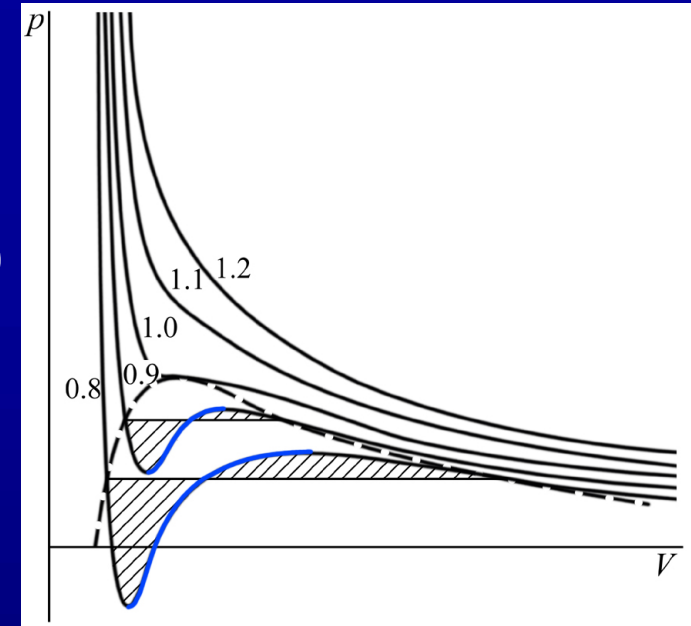
$$\tilde{p} \equiv \frac{p}{p_C}, \quad \dots$$

Problems:

1. non-physical part ($p < 0$)
2. There is no solid phase.
3. Stability requires

$$\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T > 0$$

Positive slope part is not stable.



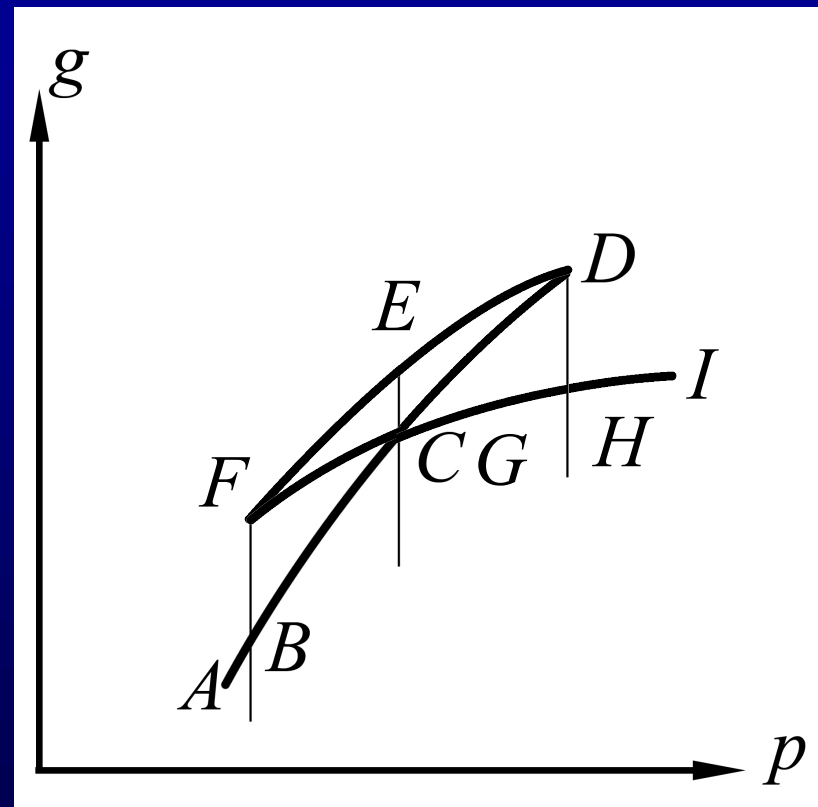
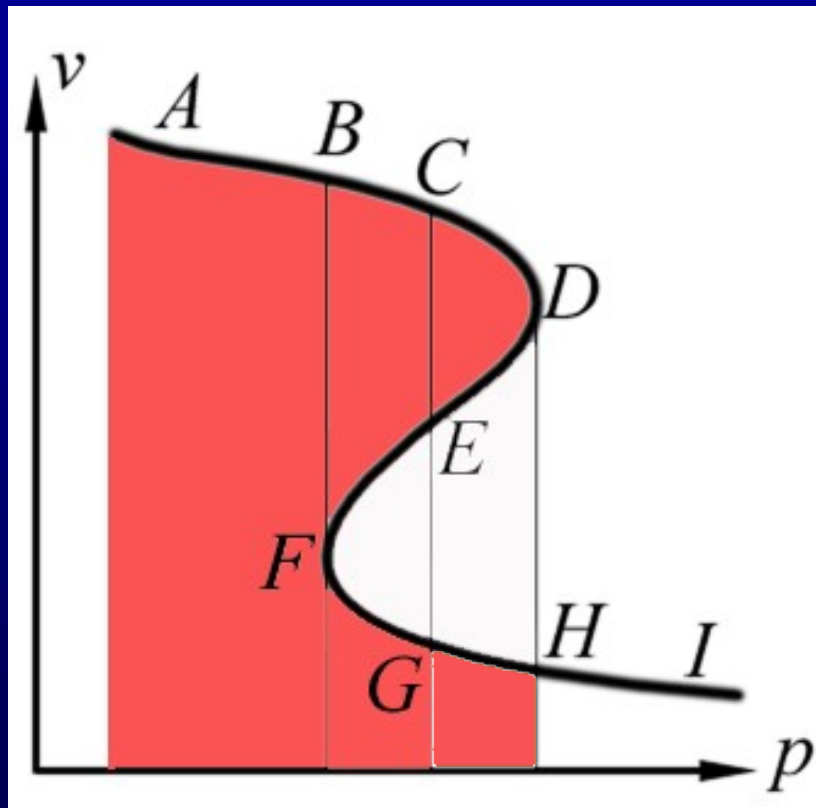
Maxwell construction & metastable states

$$dG = -SdT + Vdp + \mu dn$$

For **one mole** substance along **isotherms**, we have

$$dg = vdp$$

$$g = \int v dp$$



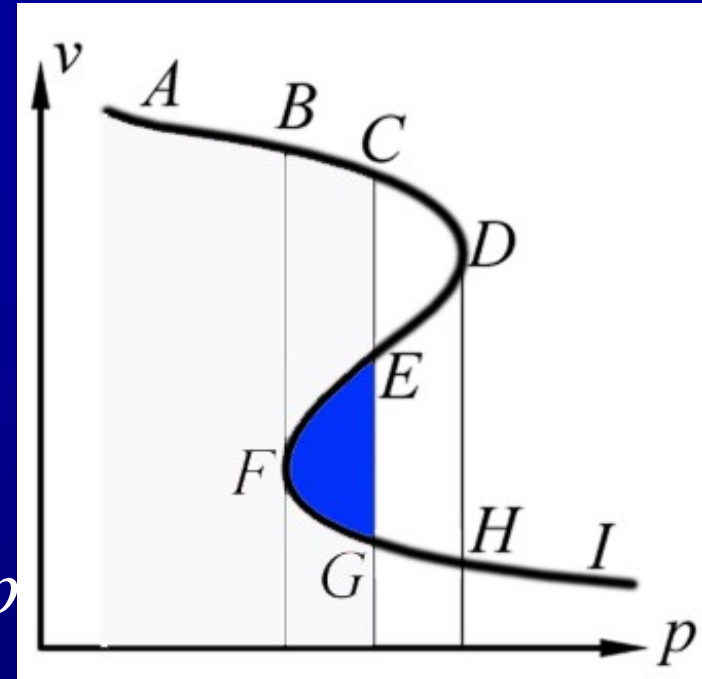
$$0 = g_G - g_C$$

$$= \int_{p_C}^{p_G} v(p) dp$$

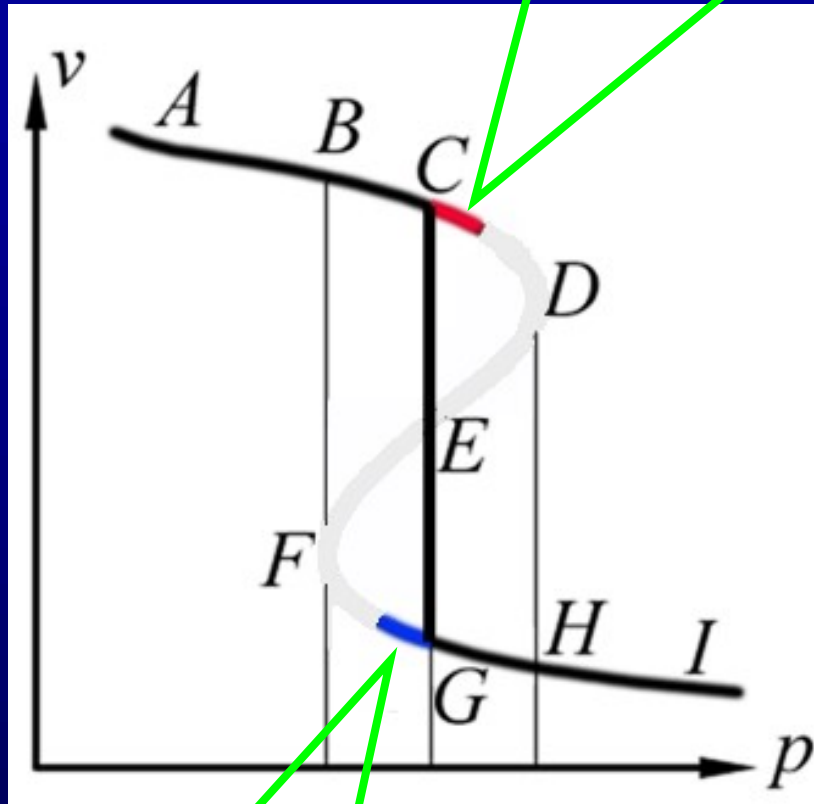
$$= \left(\int_{p_C}^{p_D} + \int_{p_D}^{p_E} + \int_{p_E}^{p_F} + \int_{p_F}^{p_G} \right) v(p) dp$$

$$= \left(\int_{p_C}^{p_D} - \int_{p_E}^{p_D} \right) - \left(\int_{p_F}^{p_E} - \int_{p_F}^{p_G} \right) v(p) dp$$

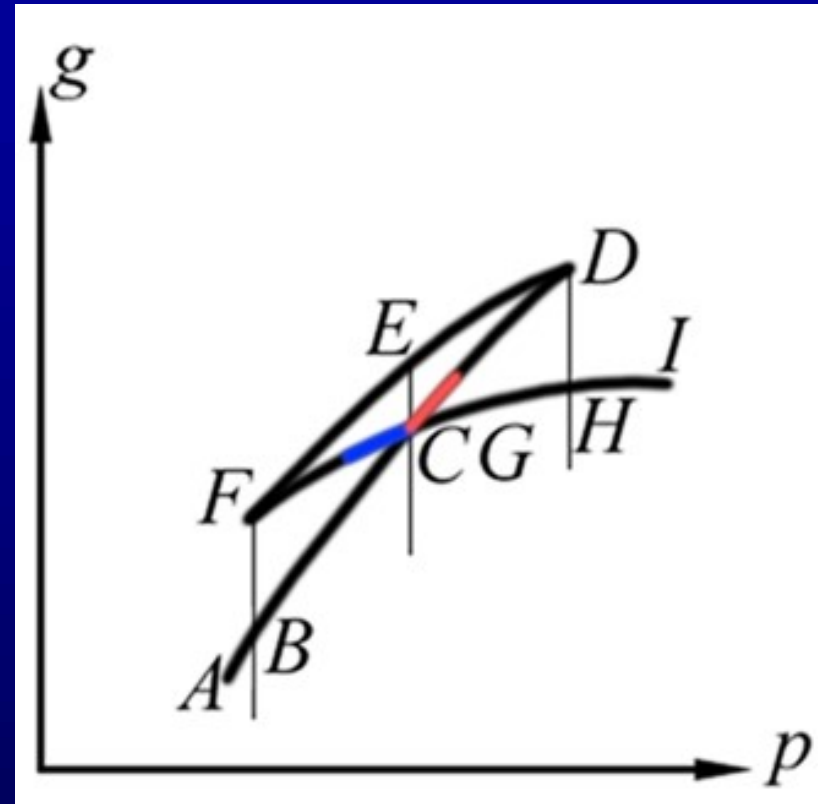
$$= A_{CDEC} - A_{EFGE}$$

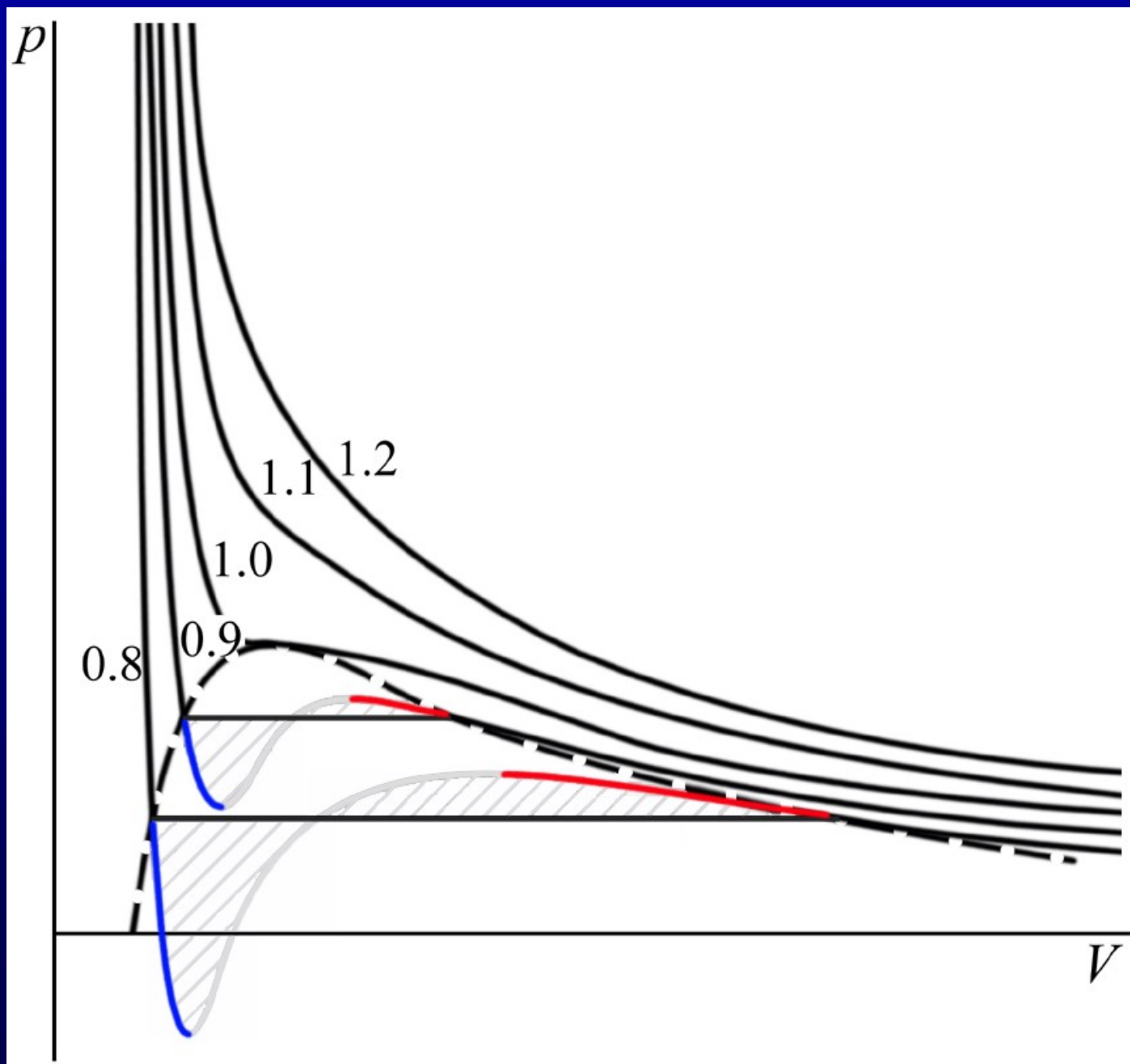


supersaturated vapor

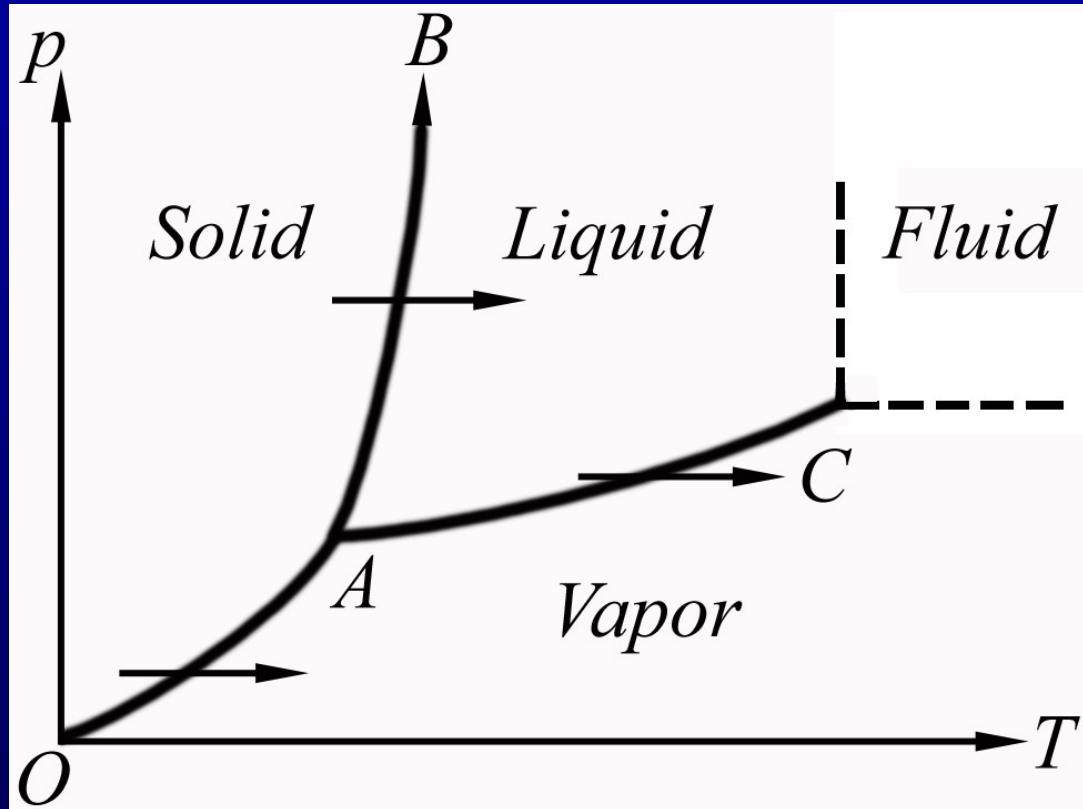


superheated liquid





14.2 Phase and phase diagrams



sublimation
curve (OA),
fusion curve
(AB),
vaporization
curve (AC)

Metastable states

Points: A , C , and B

Supercritical fluid

solid $\rightarrow$ liquid fusion, melting

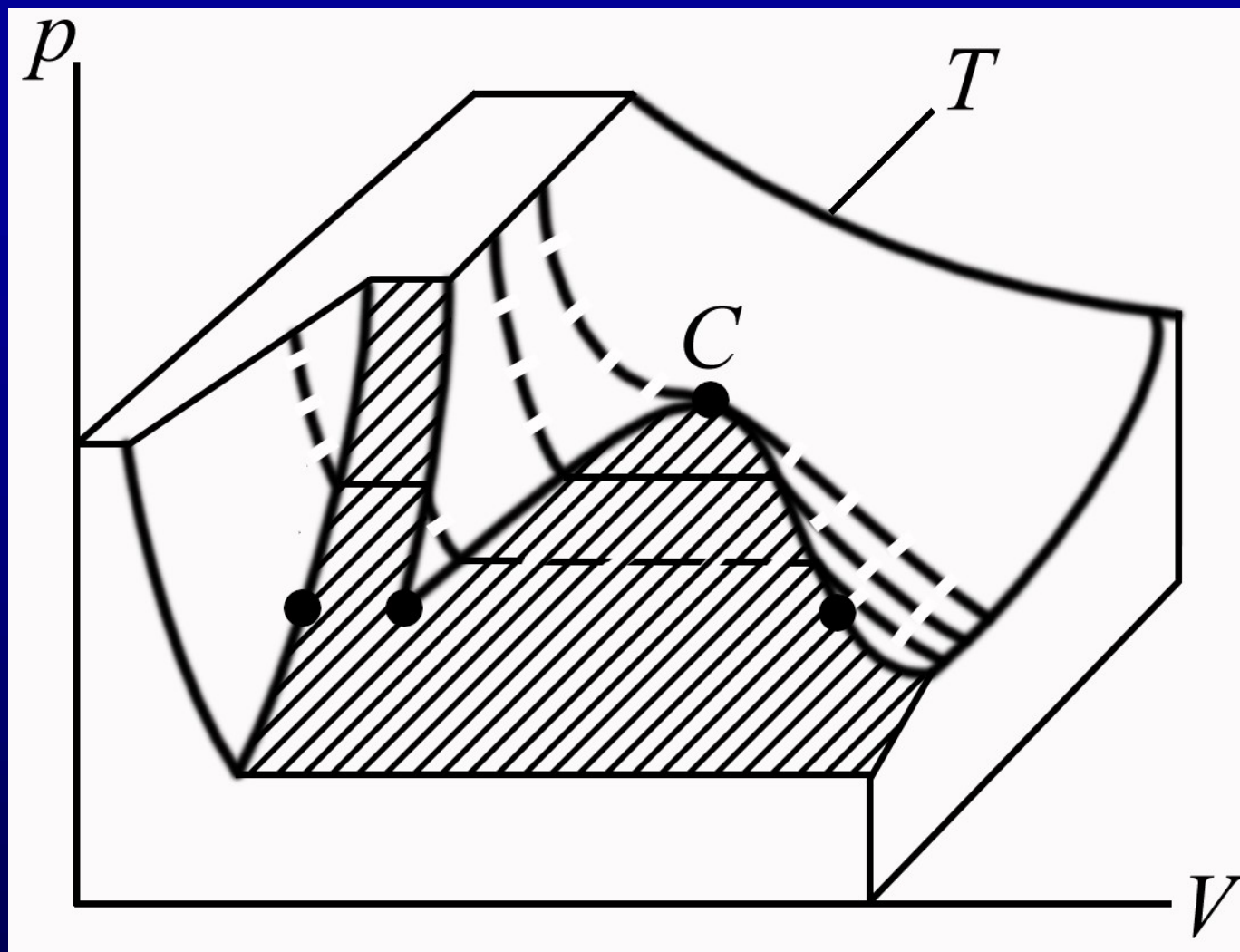
liquid $\rightarrow$ solid freezing

liquid $\rightarrow$ gas vaporization

gas $\rightarrow$ liquid condensation

solid $\rightarrow$ gas sublimation

gas $\rightarrow$ solid deposition



$$f(p, V, T) = 0$$

14.3 Clapeyron's equation

Along phase equilibrium curve

$$\mu^{\text{I}}(T, p) = \mu^{\text{II}}(T, p)$$

$$d\mu^{\text{I}} = d\mu^{\text{II}}$$

According to Gibbs-Duheim equation

$$-S_{\text{m}}^{\text{I}}dT + V_{\text{m}}^{\text{I}}dp = -S_{\text{m}}^{\text{II}}dT + V_{\text{m}}^{\text{II}}dp$$

Along coexistence curve

$$\left(\frac{\partial p}{\partial T}\right)_{\text{co}} = \frac{S_{\text{m}}^{\text{I}} - S_{\text{m}}^{\text{II}}}{V_{\text{m}}^{\text{I}} - V_{\text{m}}^{\text{II}}}$$

From $G = H - TS$ $\mu = H_m - TS_m$

Crossing the phase boundary $0 = \Delta H_m - T\Delta S_m$

$$\left(\frac{dp}{dT}\right)_{co} = \frac{\Delta H_m}{T\Delta V_m} = \frac{\ell}{T\Delta V_m}$$

Latent heat

The **slope** of vaporization curve

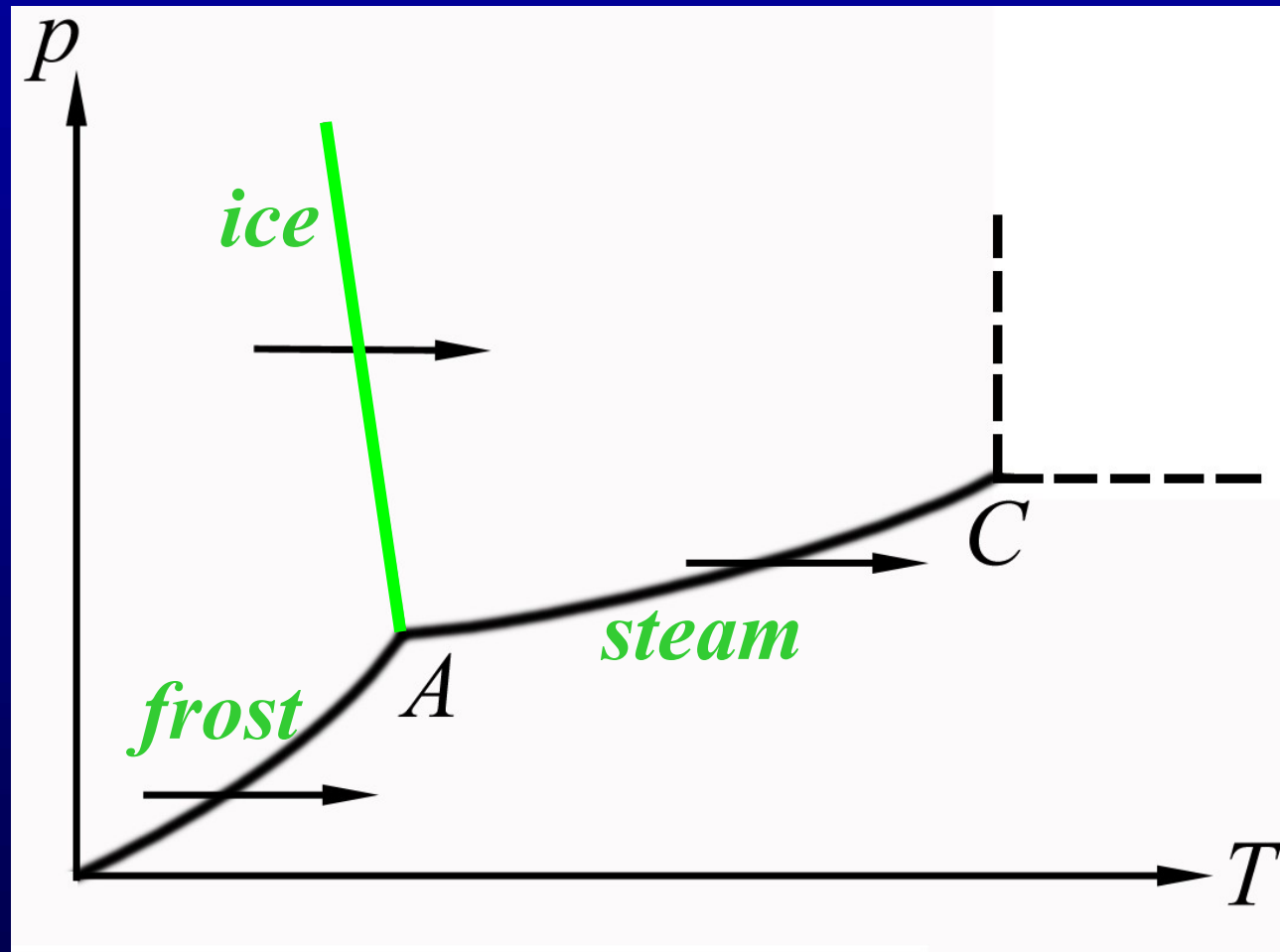
$$V_{mL} \ll V_{mV} = \frac{RT}{p}$$

$$\frac{dp}{dT} = \frac{\ell_v p}{RT^2}$$

Near boiling point T_b

$$\ln \frac{p}{p_C} = C - \frac{\ell_{vb}}{RT}$$

phase diagram of water



$$\left(\frac{\partial p}{\partial T} \right)_{\text{co}} = \frac{S_{\text{m}}^{\text{I}} - S_{\text{m}}^{\text{II}}}{V_{\text{m}}^{\text{I}} - V_{\text{m}}^{\text{II}}}$$

ordered

The ice point is equilibrium temperature when

- **air + steam**
- **ice**
- **water + air**

co-exist under atmospheric pressure.

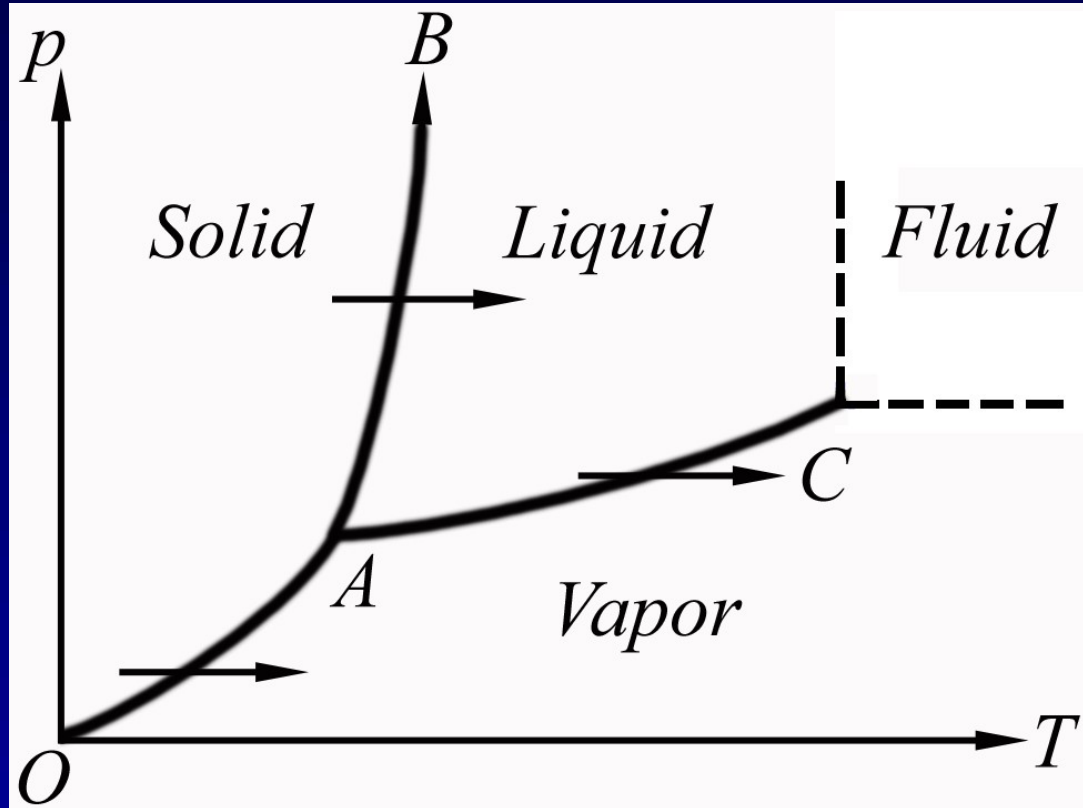
$$\begin{aligned}
\Delta T &= T - T_{\text{tr}} & \left(\frac{dp}{dT} \right)_{\text{co}} &= \frac{\Delta H_{\text{m}}}{T \Delta V_{\text{m}}} = \frac{\ell}{T \Delta V_{\text{m}}} \\
&= \frac{T(V_{\text{mw}} - V_{\text{mi}})}{\ell_{\text{F}}} \Delta p \\
&= \frac{273\text{K} \times (1.00 \times 10^{-3} - 1.09 \times 10^{-3}) \text{m}^3/\text{kg}}{3.34 \times 10^5 \text{J/kg}} \\
&\quad \times (760\text{mmHg} - 4.584\text{mmHg}) \times \frac{101325\text{Pa}}{760\text{mmHg}} \\
&\approx -0.0075\text{K}
\end{aligned}$$

**Air solution contributes an additional –
0.0023K**

$$T_{\text{ice}} = T_{\text{tr}} - 0.0098\text{K} \approx 273.15\text{K}$$

Assignment: 14.2, *14.5

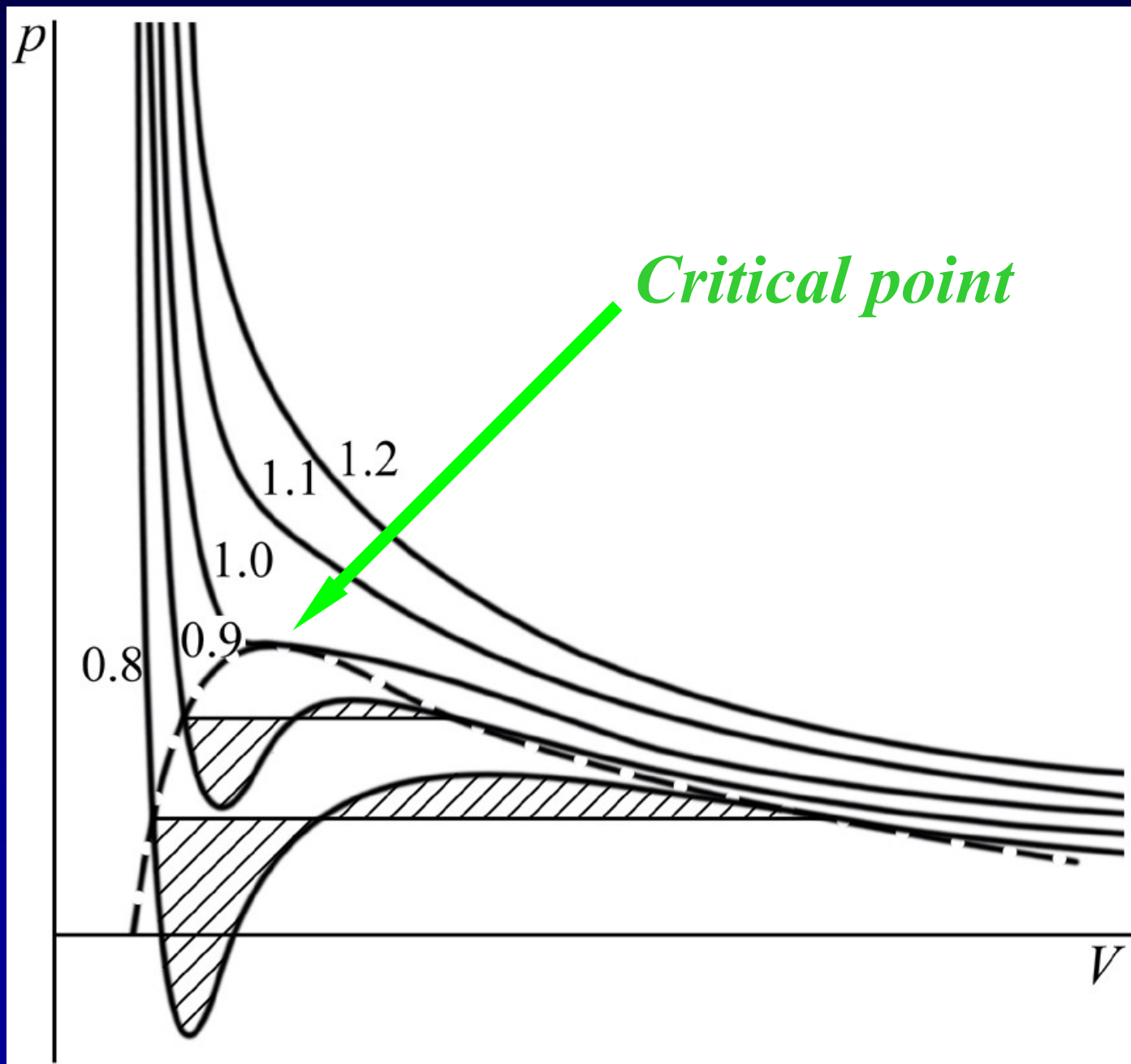
14.2 Phase and phase diagrams



sublimation
curve (OA),
fusion curve
(AB),
vaporization
curve (AC)

Metastable states

Points: A , C , and B



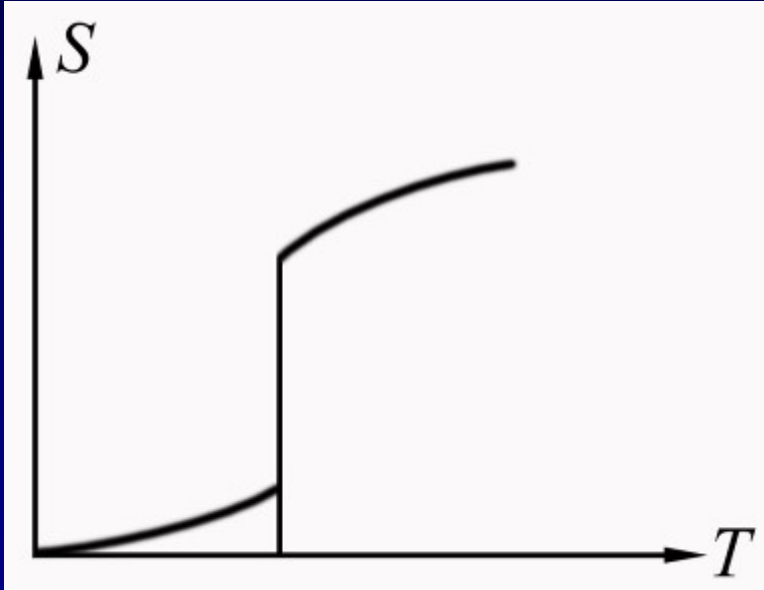
14.4 Higher order phase transition

Ehrenfest (1933) scheme:

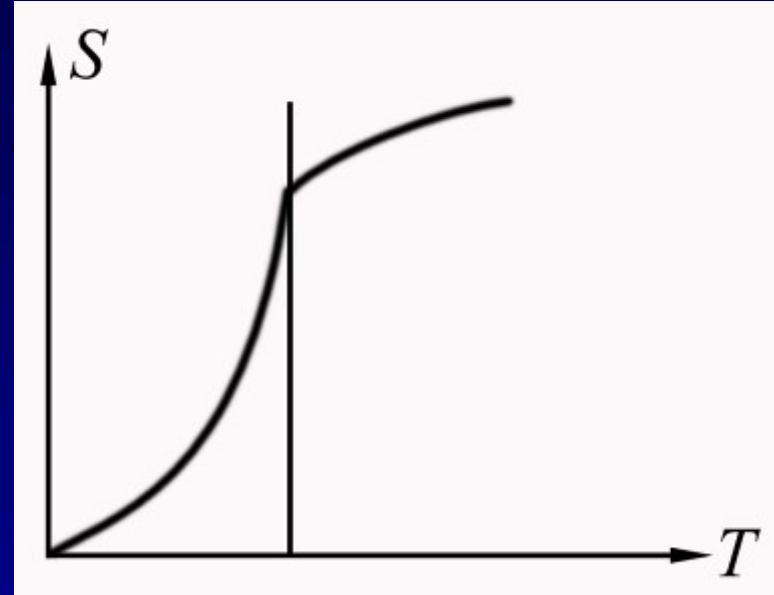
a phase transition is n th order if n th order derivative of chemical potential μ is the **first discontinuous one**.

$$S_m = -\left(\frac{\partial\mu}{\partial T}\right)_{n,p}, \quad V_m = \left(\frac{\partial\mu}{\partial p}\right)_{n,T}$$

$$\Delta S \neq 0; \quad \Delta V \neq 0 \longleftrightarrow \text{1st order}$$



$$\Delta S \neq 0$$



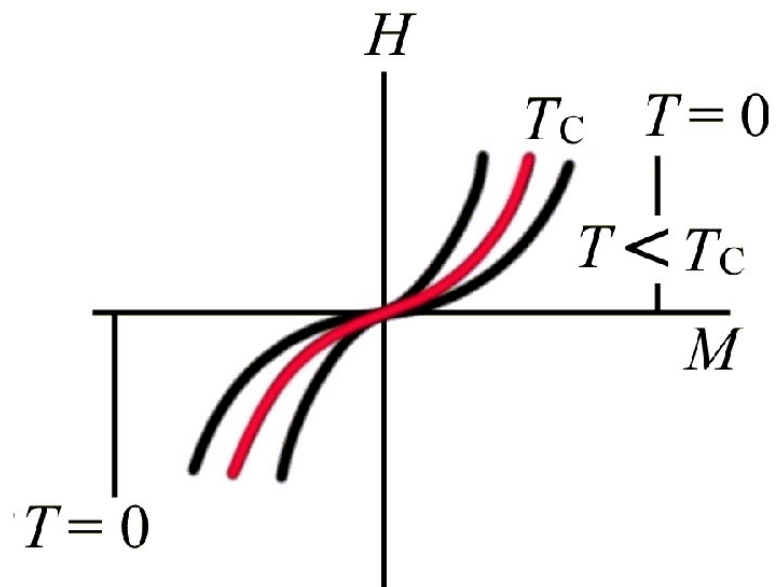
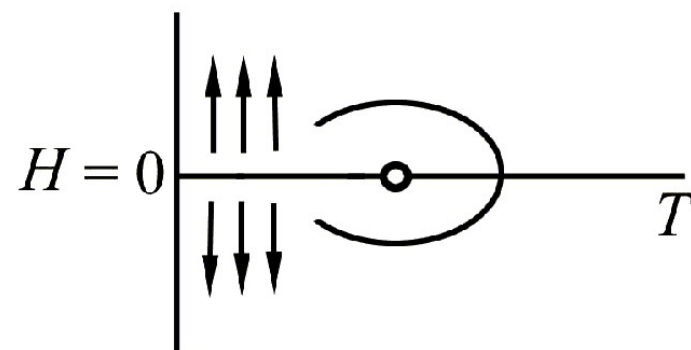
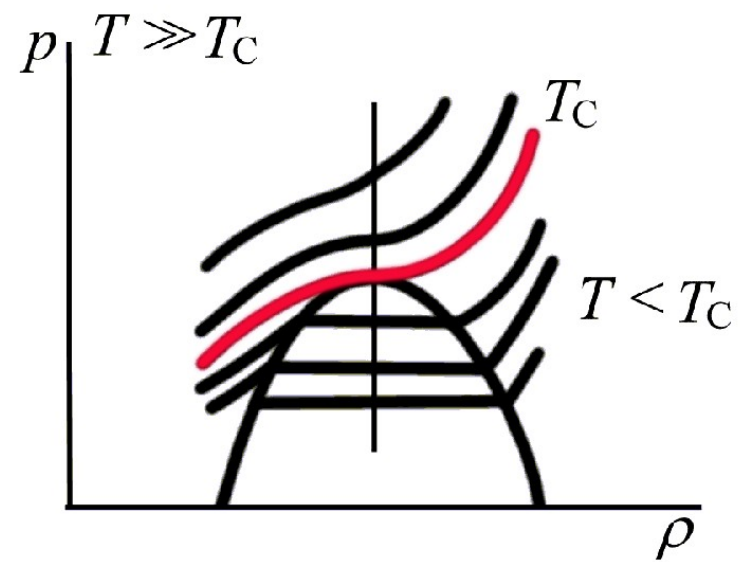
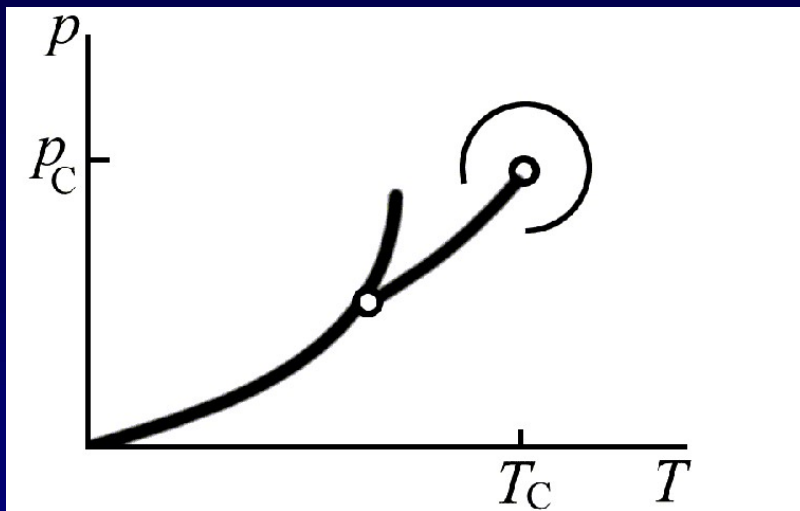
$$\Delta S = 0; \quad \Delta \left(\frac{\partial S}{\partial T} \right) \neq 0$$

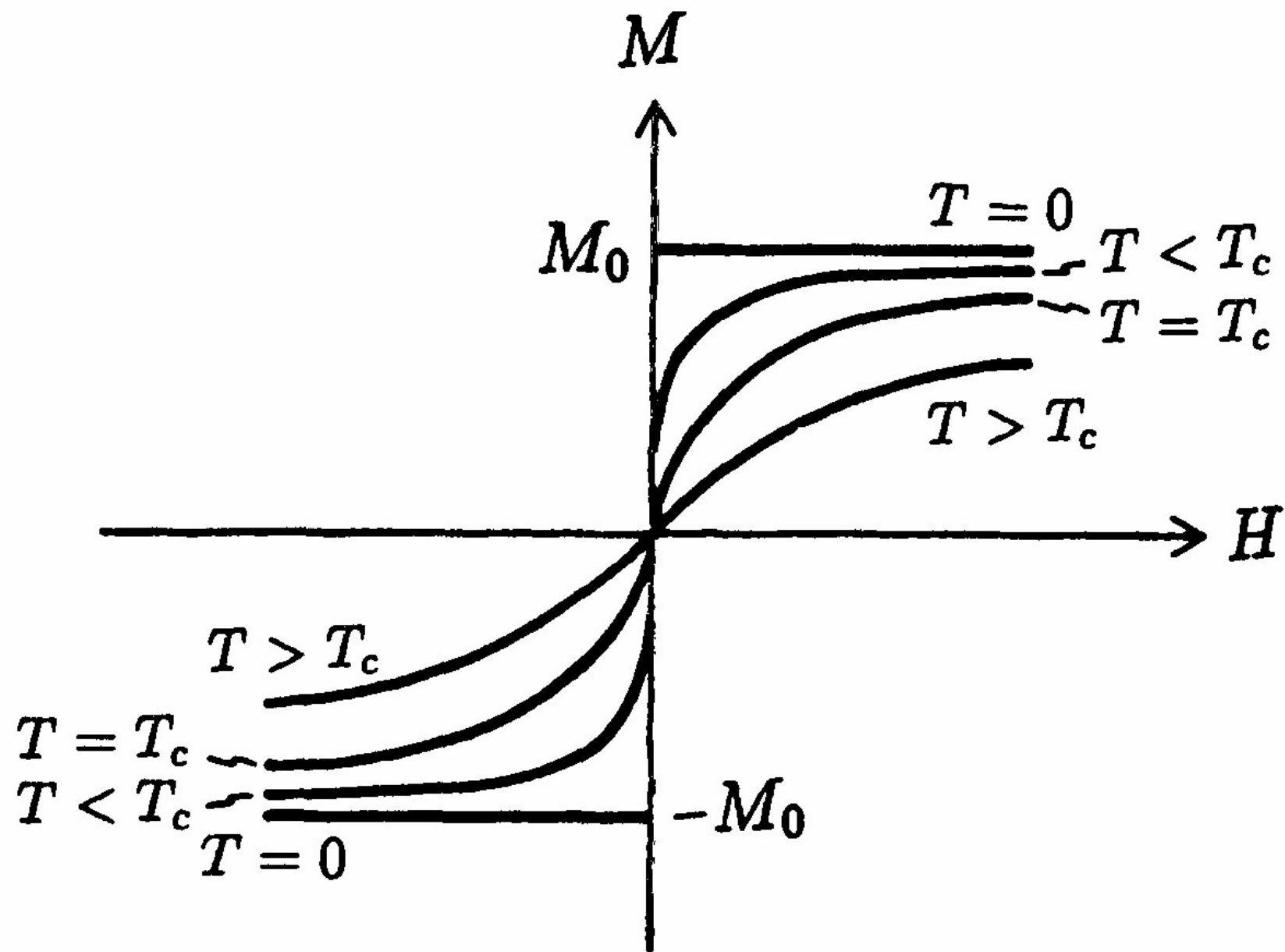
$$c \sim T \frac{dS}{dT} \sim T \frac{\partial^2 \mu}{\partial T^2}$$

$$\kappa_T \sim -\frac{1}{V} \frac{\partial V}{\partial p} \sim V \frac{\partial^2 \mu}{\partial p^2}$$

$$\chi \sim \frac{\partial M}{\partial \tilde{H}}$$

magnetic susceptibility





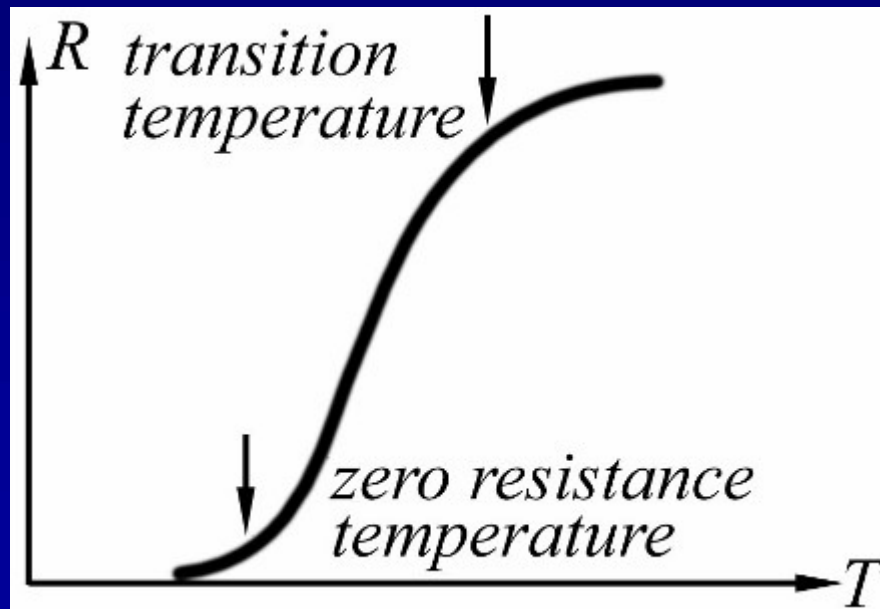
Superconductivity

1911 Onnes

1957 Bardeen-Cooper-Schrieffer

1986 Bednorz J G & Müller K A.

(oxide superconductor $T_C \sim 134\text{K}$)

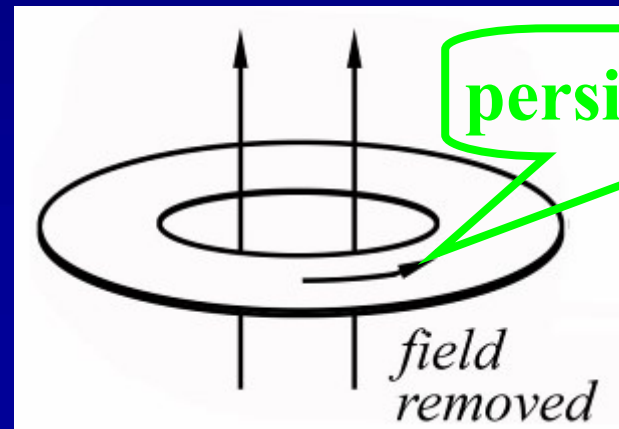
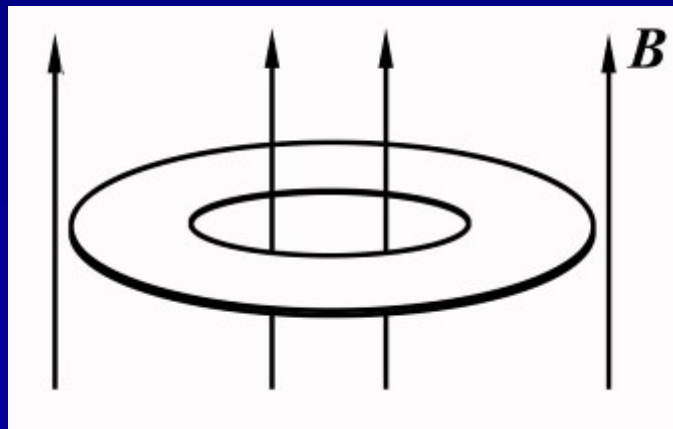
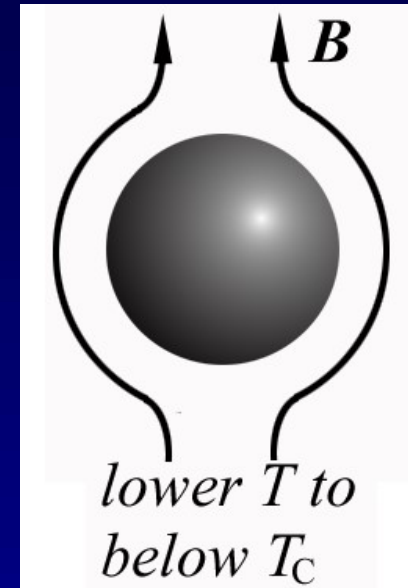
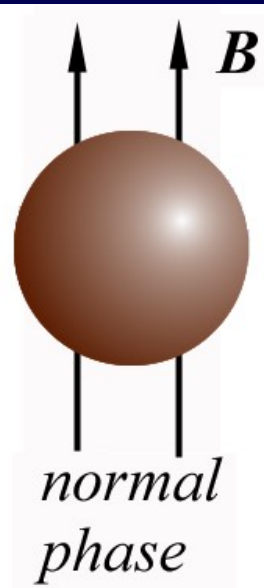
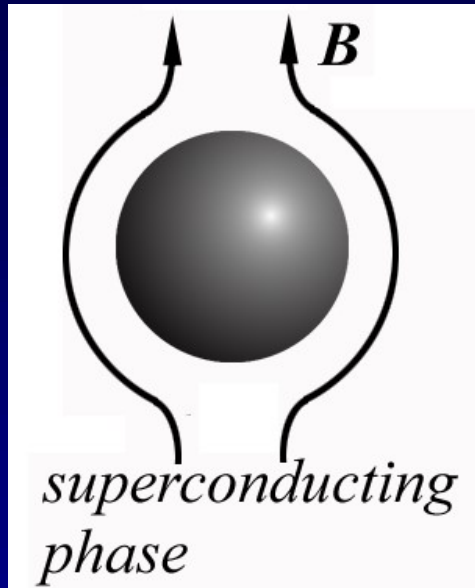


$$-\frac{1}{C} \frac{\partial B}{\partial t} = \nabla \times E$$

$$\nabla \times \left(\frac{j}{\sigma} \right) \xrightarrow{\sigma \rightarrow \infty} B = C$$

Infinite conductivity

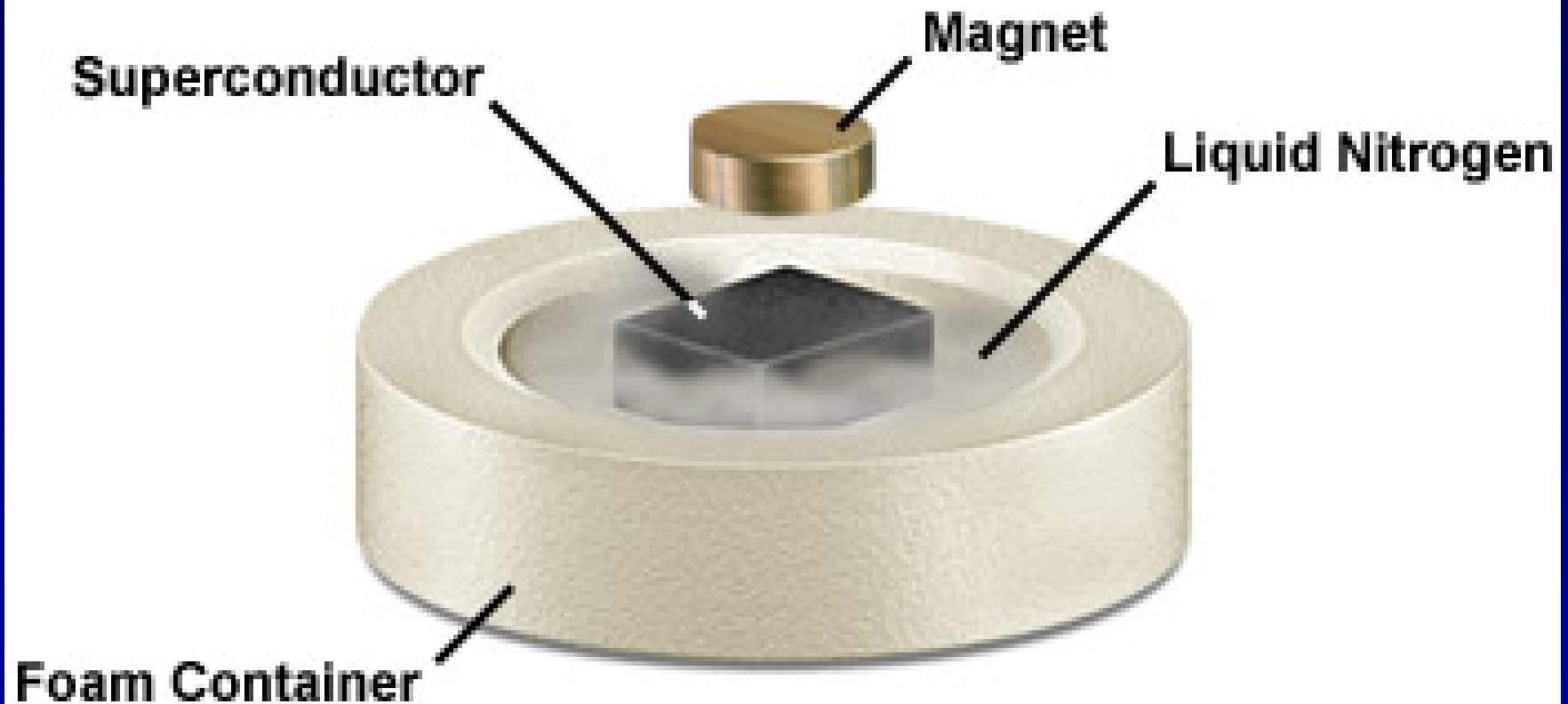
Meissner effect (perfect diamagnetism)

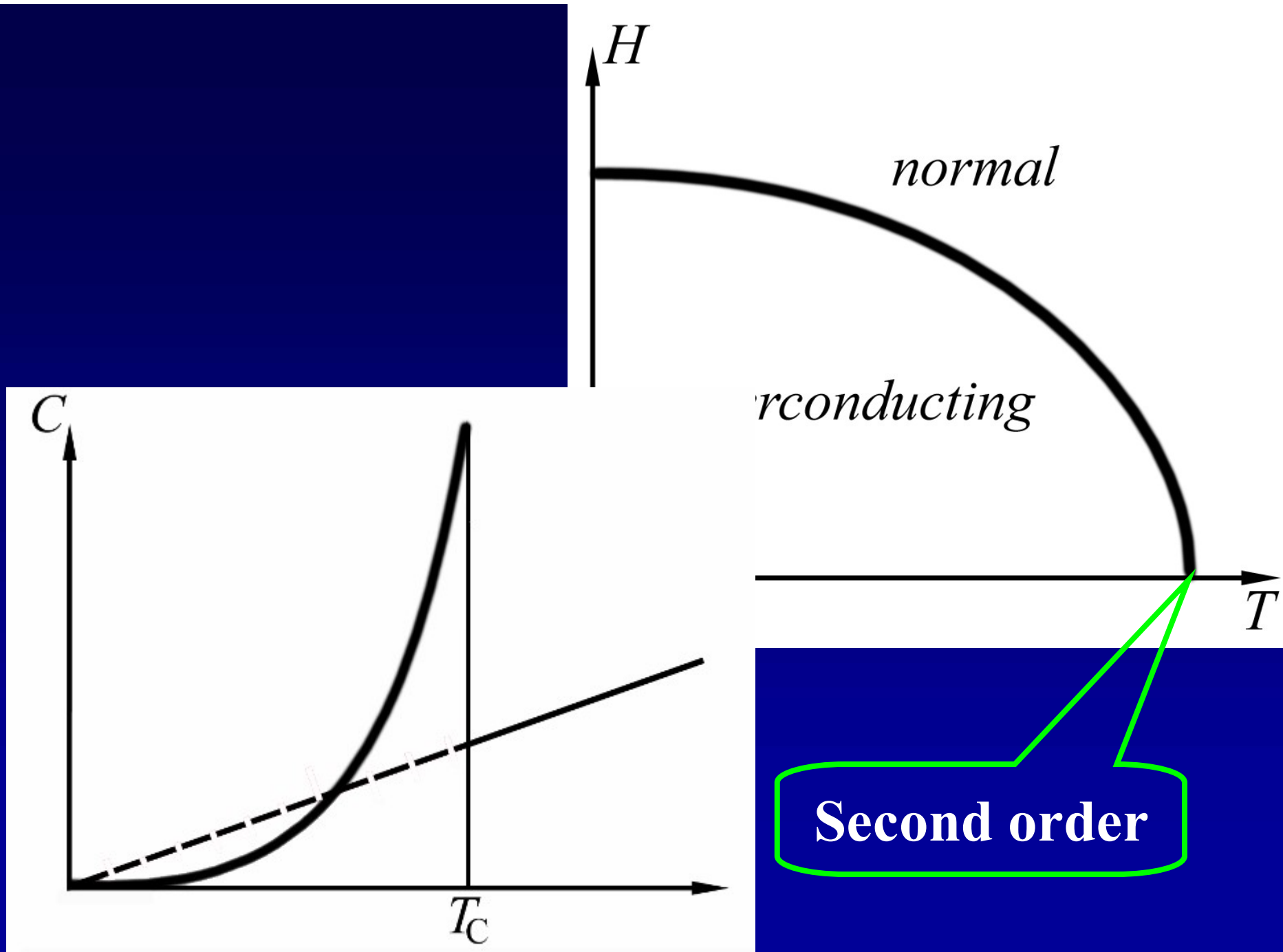


persistent current

magnetic levitation

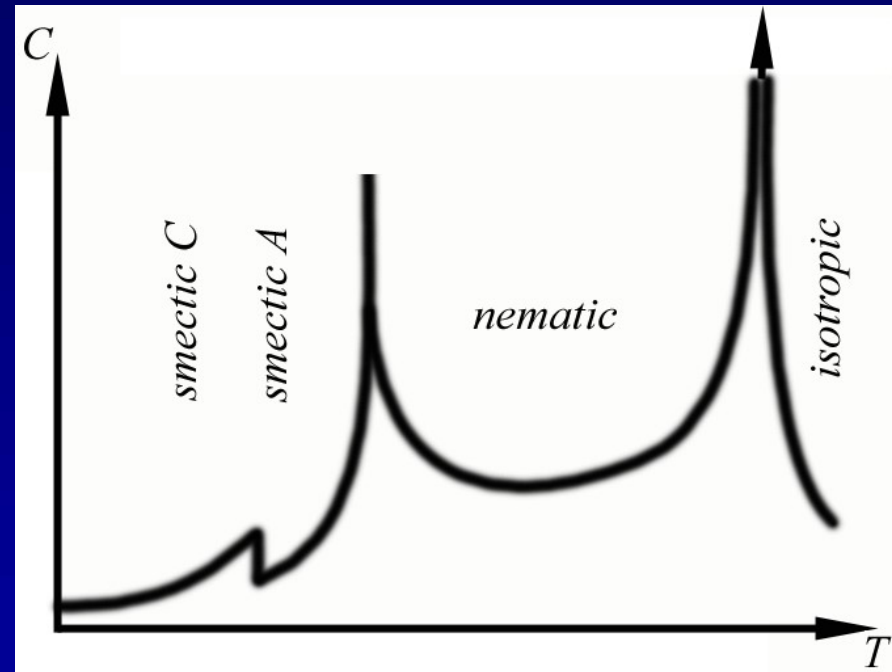
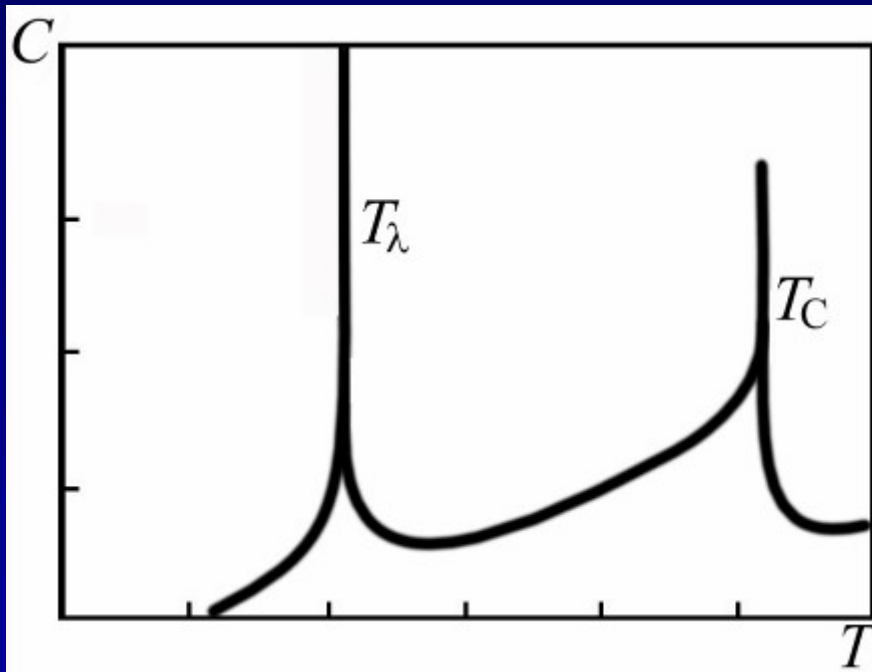
The Meissner Effect





Problems of the scheme:

- In some cases C_p , κ approach to **infinite**.
(derivative of the free energy diverges)



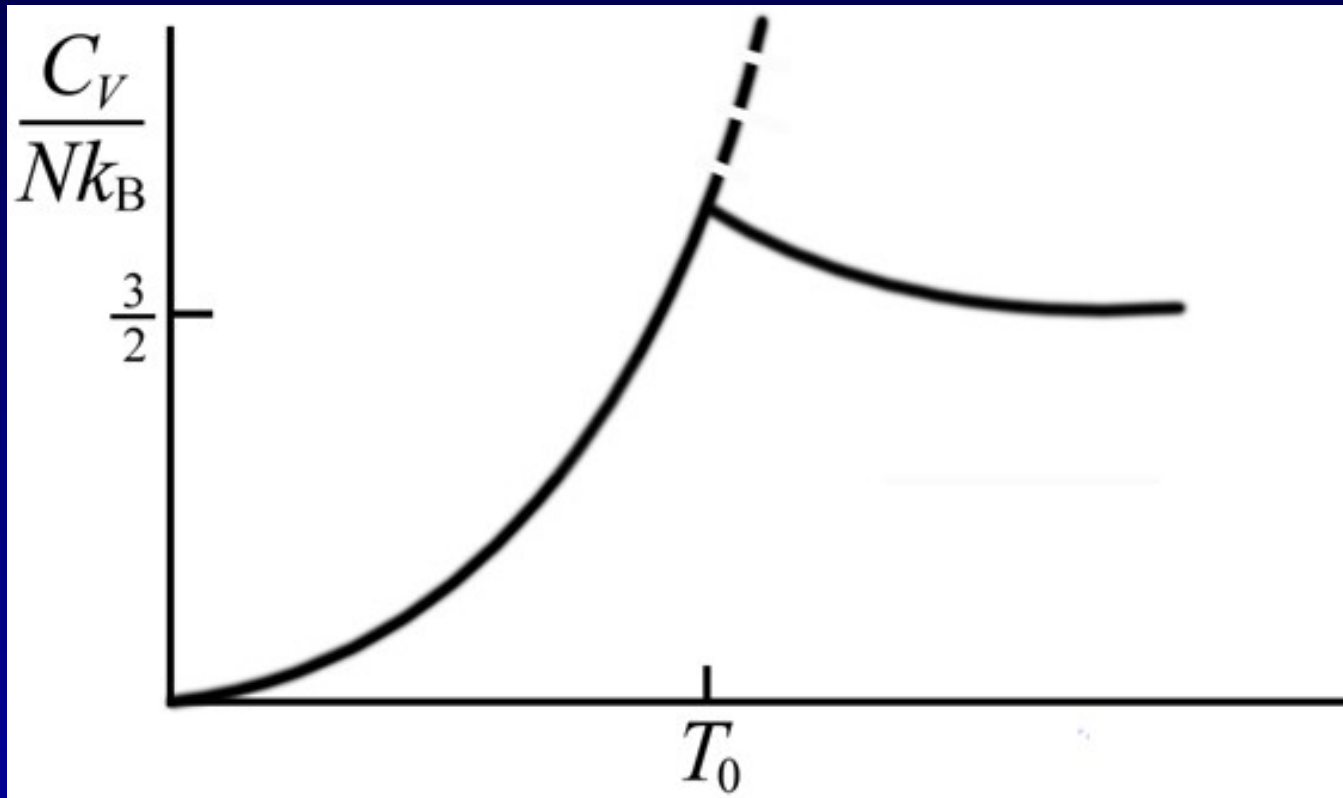
- In some phase transition (Kosterlitz-Thouless phase transition) thermodynamic function is **singular**.

$$F \propto \exp\left(-\frac{b}{t^{1/2}}\right) \quad \left(t \equiv \frac{T - T_C}{T_C}\right)$$

essential singularity

- **Few** phase transitions are higher than second order.

C is continuous



Bose-Einstein condensation: a phase transition of higher order (third order)

Modern classification of phase transitions

first order phase transitions

(have associated latent heat
mixed-phase regimes)

continuous (higher order) phase transitions

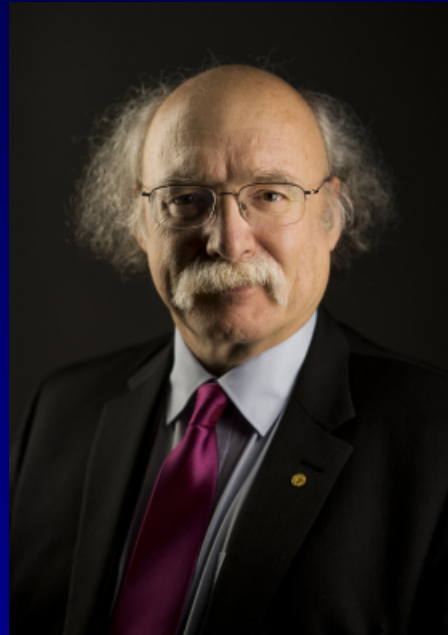
(have associated latent heat)

Critical phenomenon is indeed a transition of second order.

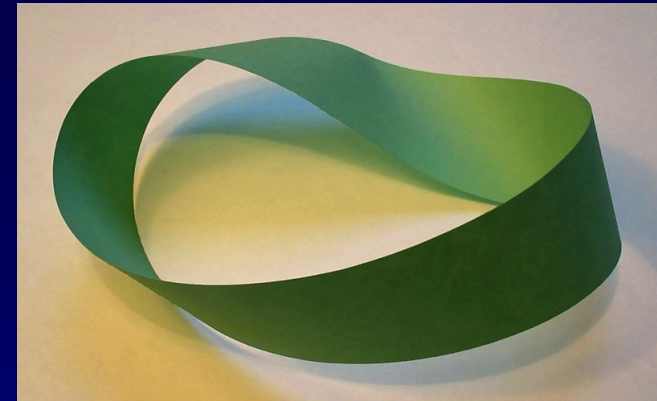
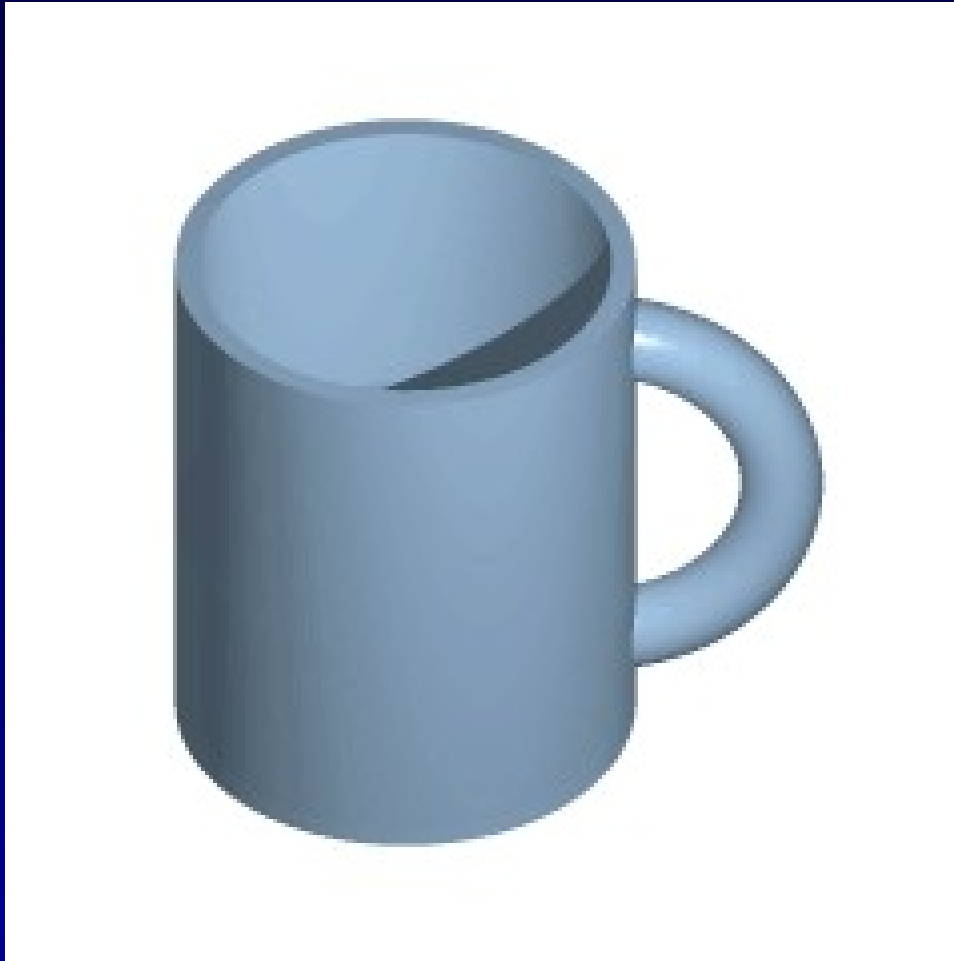
(associated with the phenomenon of **critical opalescence, a milky appearance of the liquid, due to density fluctuations at all possible wavelengths)**

Several transitions are known as the infinite-order phase transitions. The most famous example is the Kosterlitz-Thouless transition in the two-dimensional XY model.

Topological Phase Transition



The Nobel Prize in Physics 2016 was awarded with one half to David J. Thouless, and the other half to F. Duncan M. Haldane and J. Michael Kosterlitz "for theoretical discoveries of topological phase transitions and topological phases of matter".



Möbius strips

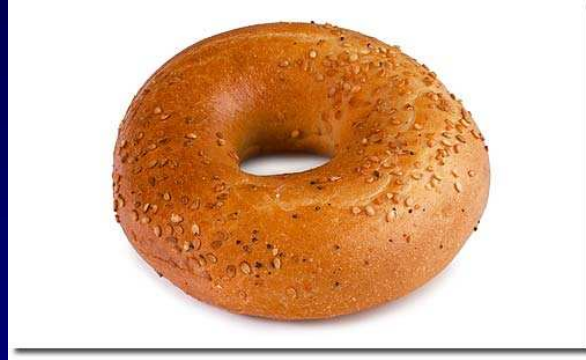
Topology is the study of qualitative properties of certain objects (called topological spaces) that are invariant under a certain kind of transformation (called homeomorphism (同胚, a bi-continuous mapping)).



cinnamon bun

肉桂面包

馒头(馍) steamed bun



bagel

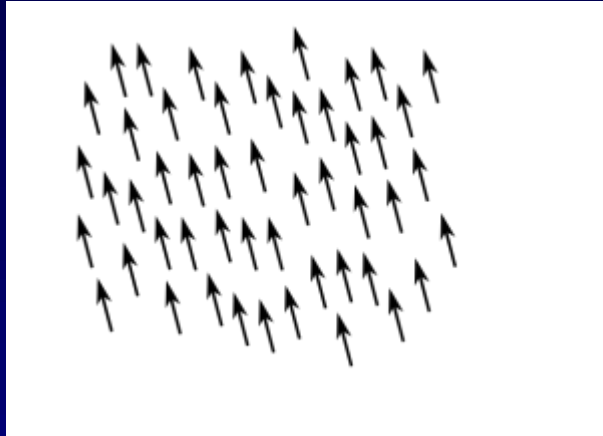
硬面包圈, “贝果面包”



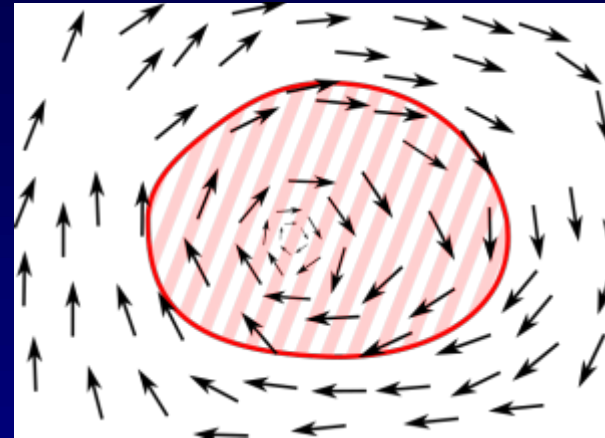
pretzel

椒盐卷饼

Brezel (German)

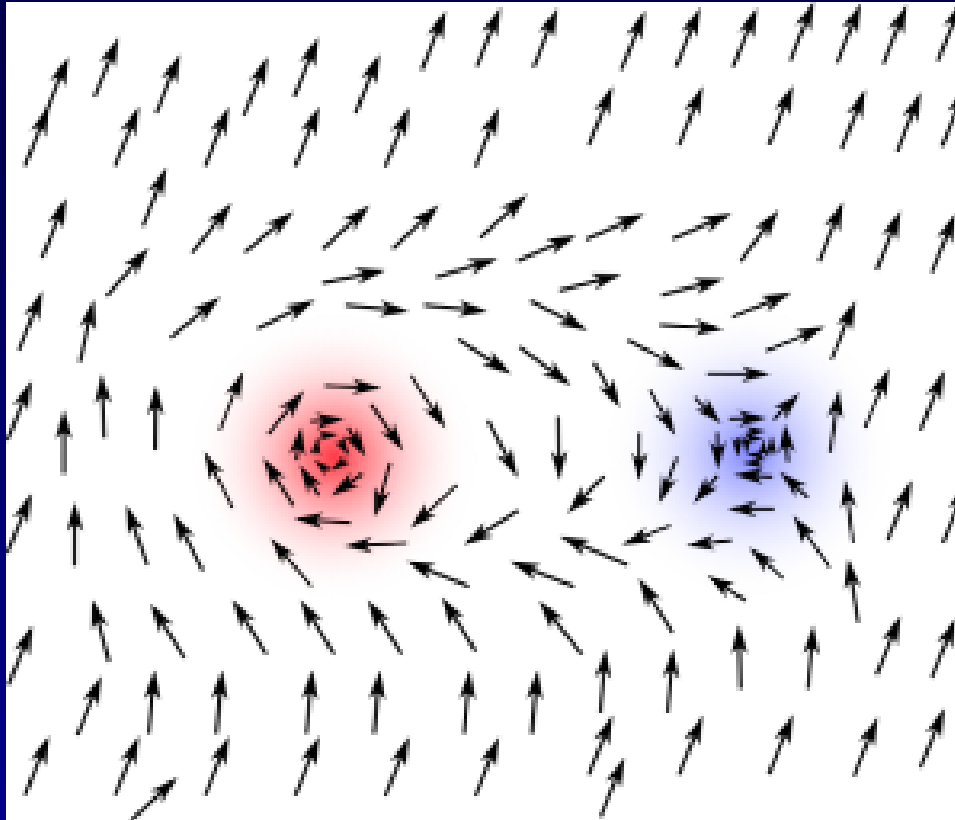


Orientation order

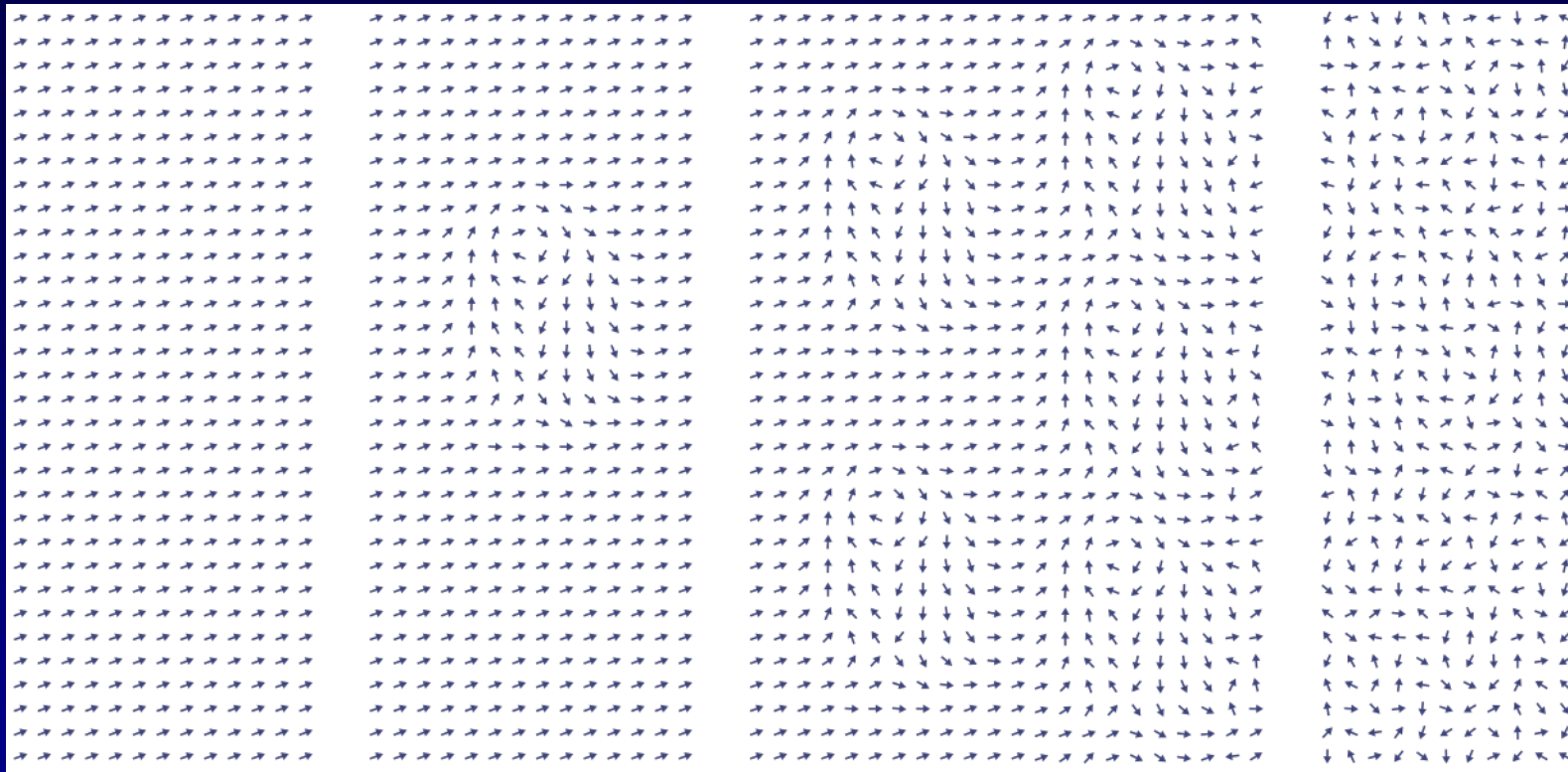


Vortex (渦旋)

**Following the red line,
the arrows wind
precisely 360 degrees
to come back to the
original direction.
winding number**



vortex–antivortex pair



1. Ordered state

2. A vortex-antivortex pair

3. Many pairs

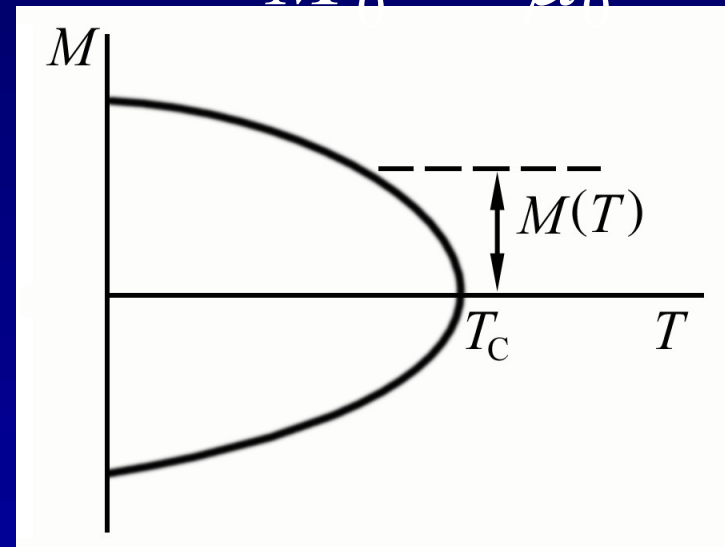
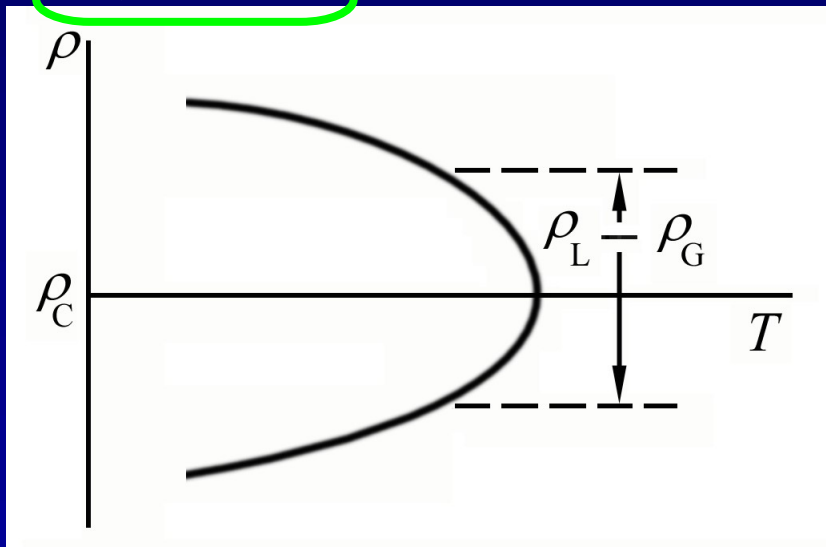
4. Vortex unbinding

*14.5 Some concepts in modern theory of phase transition

Order parameter

序参量

$$M = n \langle \mu \rangle$$
$$s = \frac{M}{M_0} = \frac{\langle \mu \rangle}{\mu_0}$$



In a continuous phase transition the order parameter changes continuously to zero.

Spontaneous symmetry breaking

more symmetrical high-temperature phase



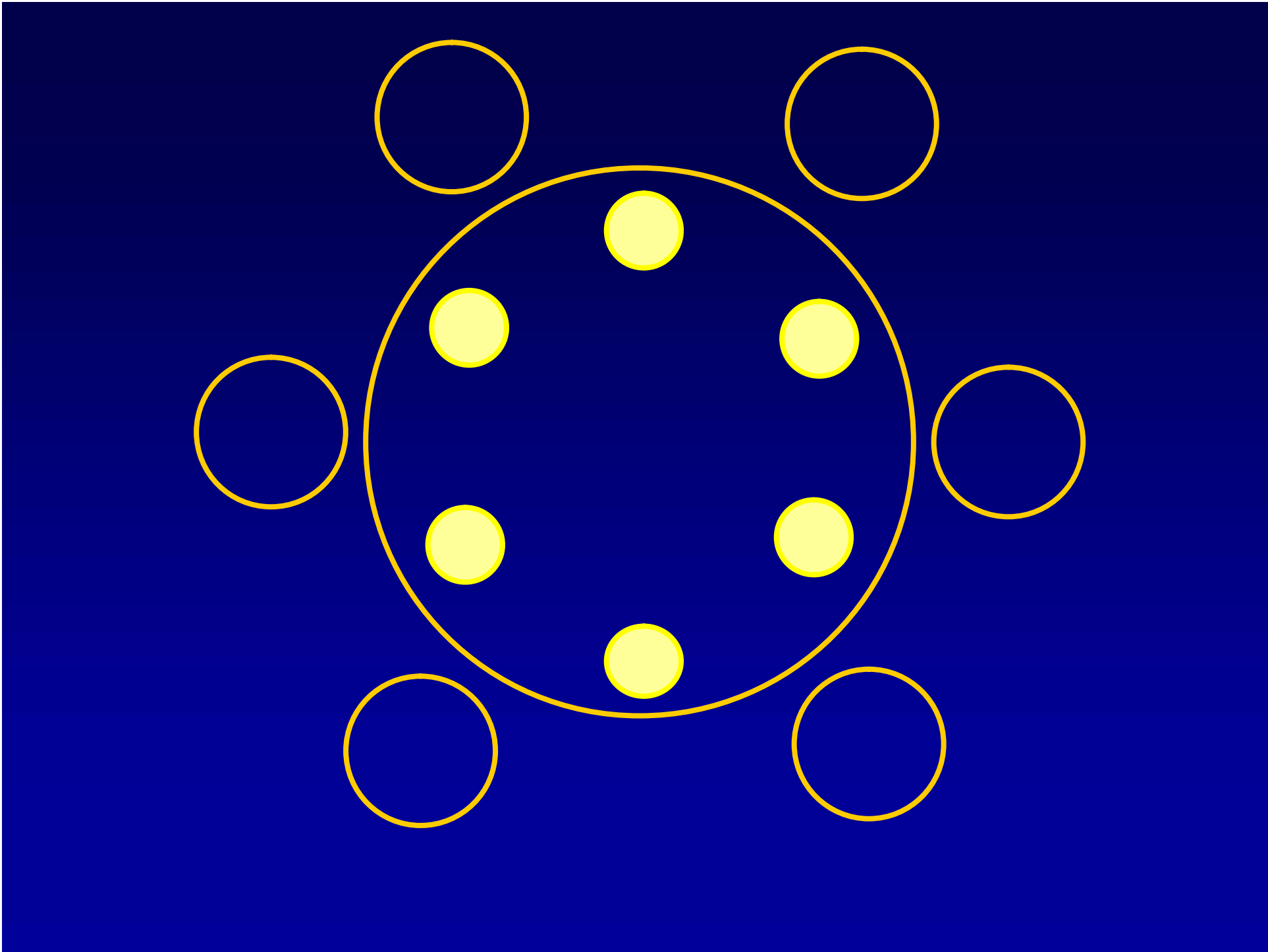
less symmetrical low-temperature phase

Paramagnetic phase:

isotropic or spherical symmetry

Ferromagnetic phase:

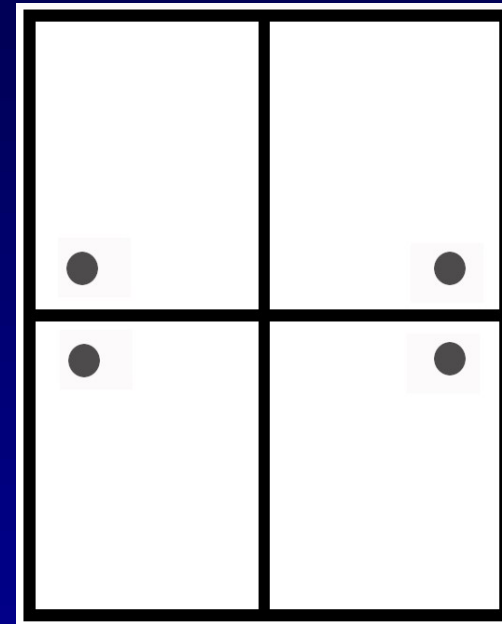
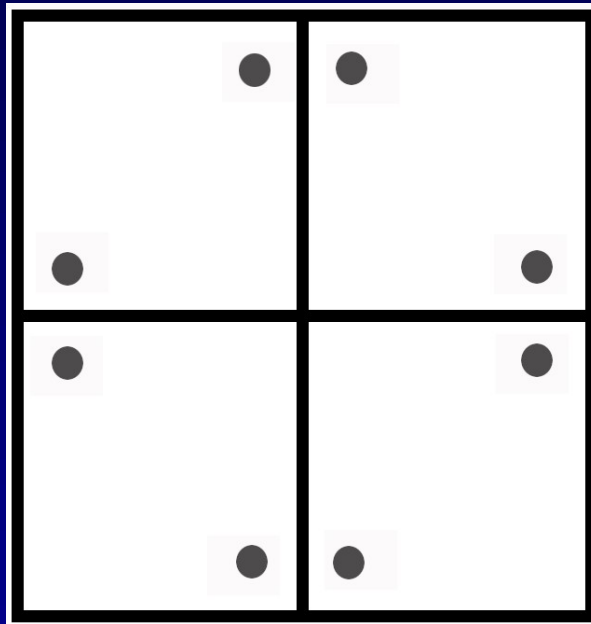
cylindrical symmetry



Continuous change in order parameter

Abrupt change in symmetry

symmetry breaking and continuous phase transition



lattice constant

$a \longrightarrow a^-$

symmetry elements 8

$\longrightarrow$ 4

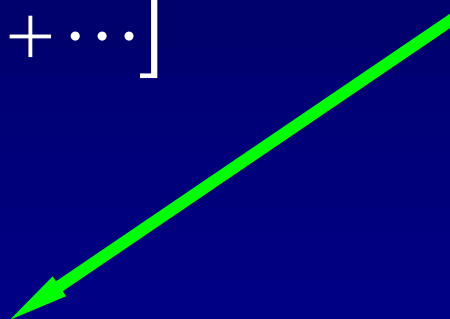
Symmetry elements can be identities, mirror planes, axes of rotation (both proper and improper), and centers of inversion.

critical exponents

$$\frac{M(T)}{M(0)} = B \left(1 - \frac{T}{T_C} \right)^\beta [1 + \dots]$$

$$\frac{\rho_L - \rho_G}{\rho_C} = B \left(1 - \frac{T}{T_C} \right)^\beta [1 + \dots]$$

1945 E. A. Guggenheim

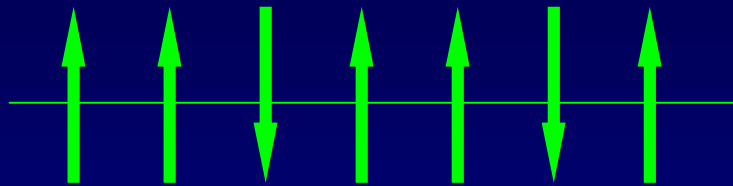
$$\frac{\rho_L - \rho_G}{\rho_C} = \frac{7}{2} \left(1 - \frac{T}{T_C} \right)^{1/3}$$


According to van der Waals equation, $\beta = 1/2$

Statistical model

1925 PhD thesis

Ising model

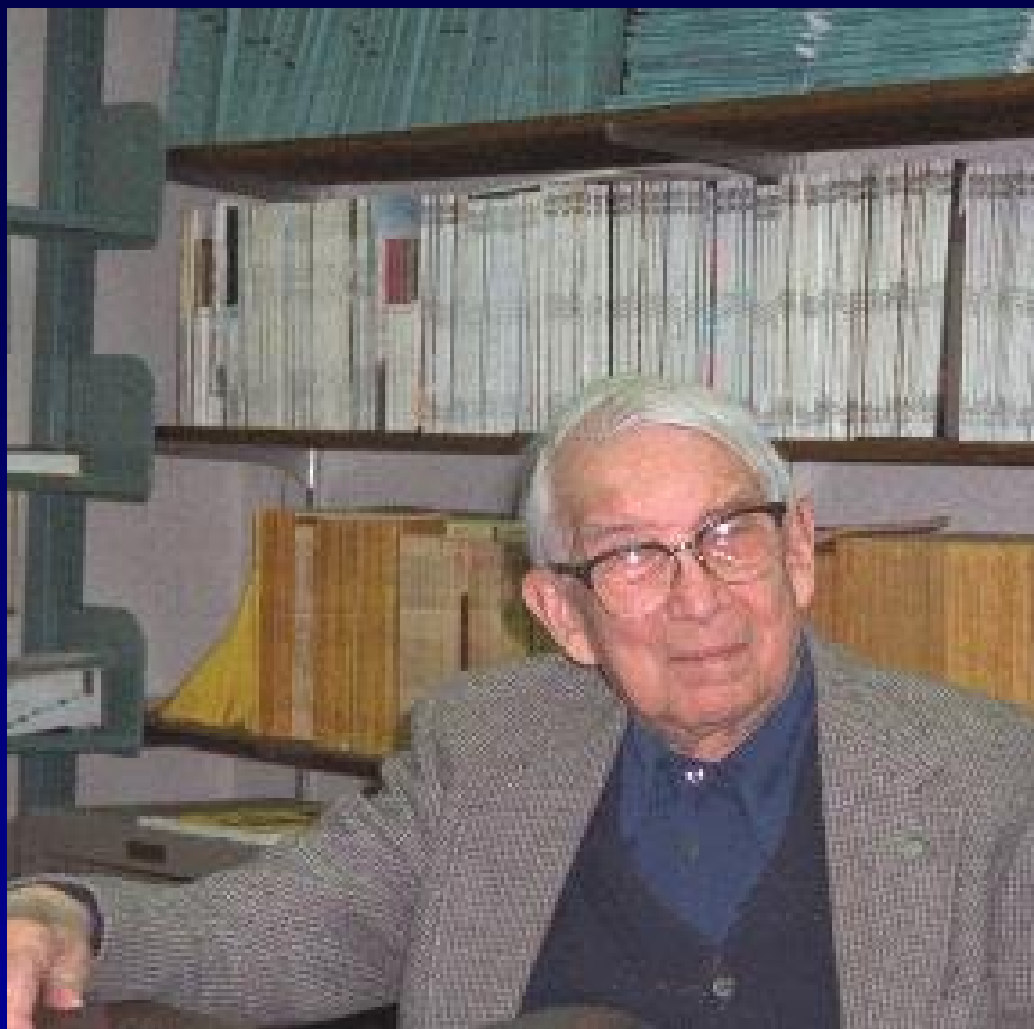


1944 Onsager found the exact solution of
2D Ising model.

1952 Yang $\beta = 1/8$

3D Ising model?

In 2000, Sorin Istrail proved that the Ising model on an arbitrary sublattice of a three dimensional square lattice is computationally intractable.



ERNEST ISING

Photograph taken in the Library of the Departments of Natural Sciences, Bradley University, Peoria(Ill.) on 11 March, 1996.

Ising model can be used in

- Neural network**
- Magnetism**
- Protein folding**
- Biological membrane**
- Social behavior**

.....

- **network or percolation;**
- **nuclear system involving quarks;**
- **universe strings;**
- **vacuum phase transition;**

Protein:

- **helix-coil transition**
- **globule-coil transition**

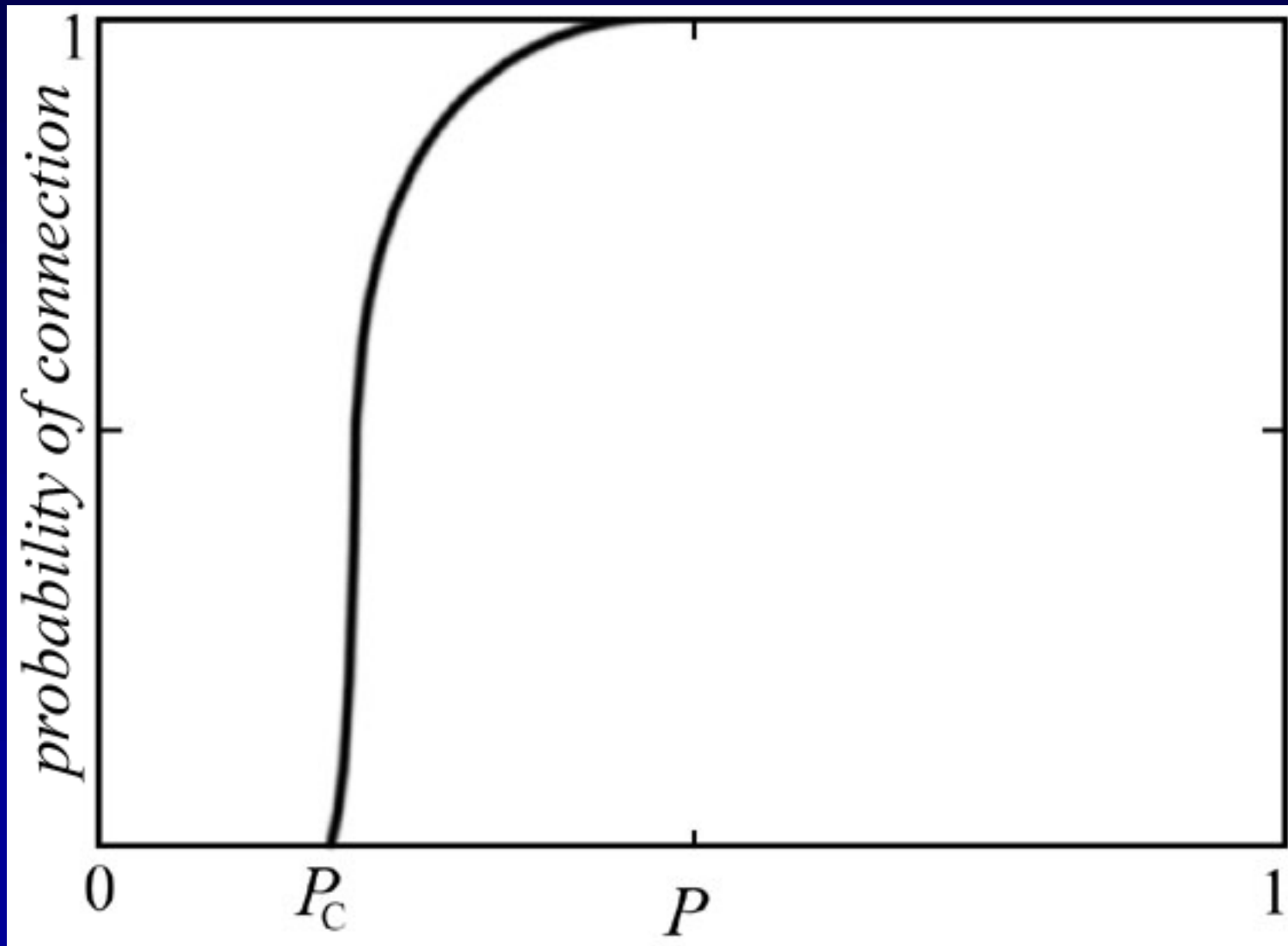
DNA:

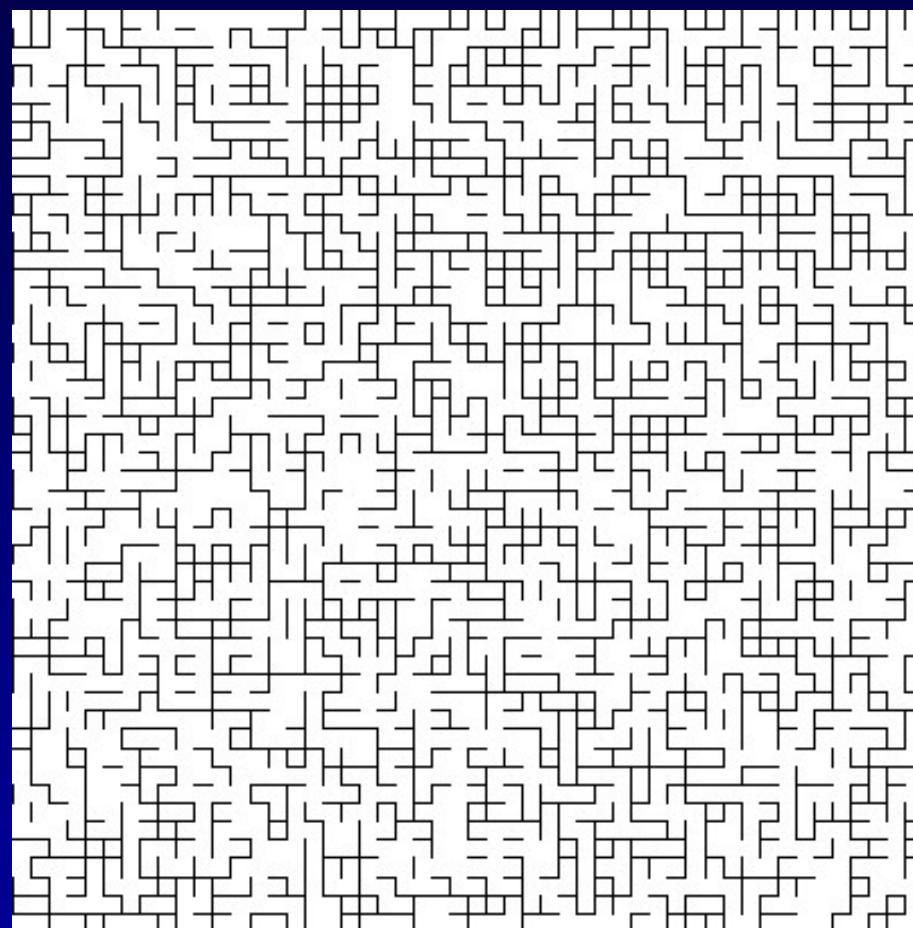
- **cooperative helix-coil transition**

- **order - chaos transition? in ion trap**

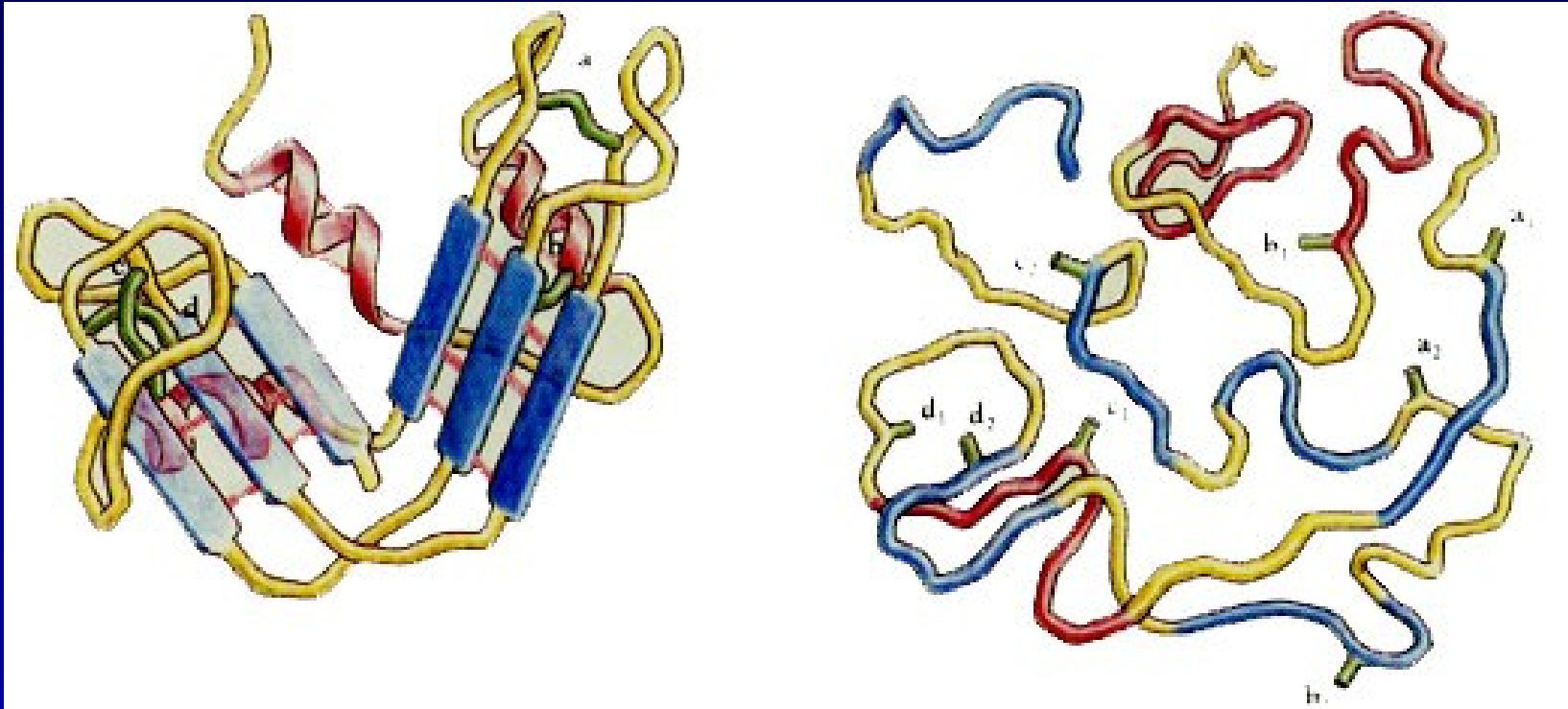
phase transition-like behaviors

percolation

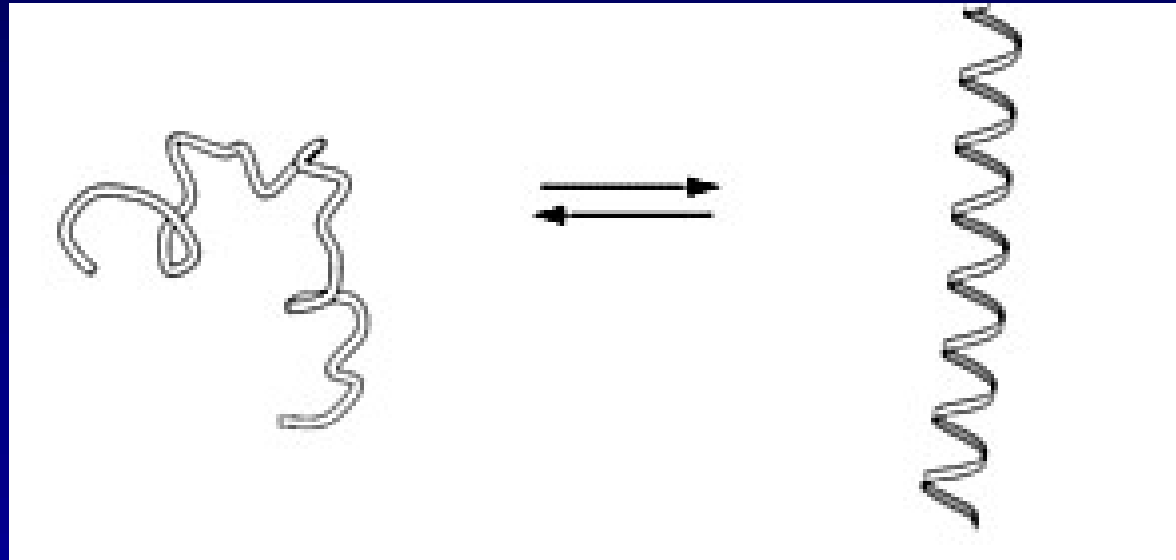




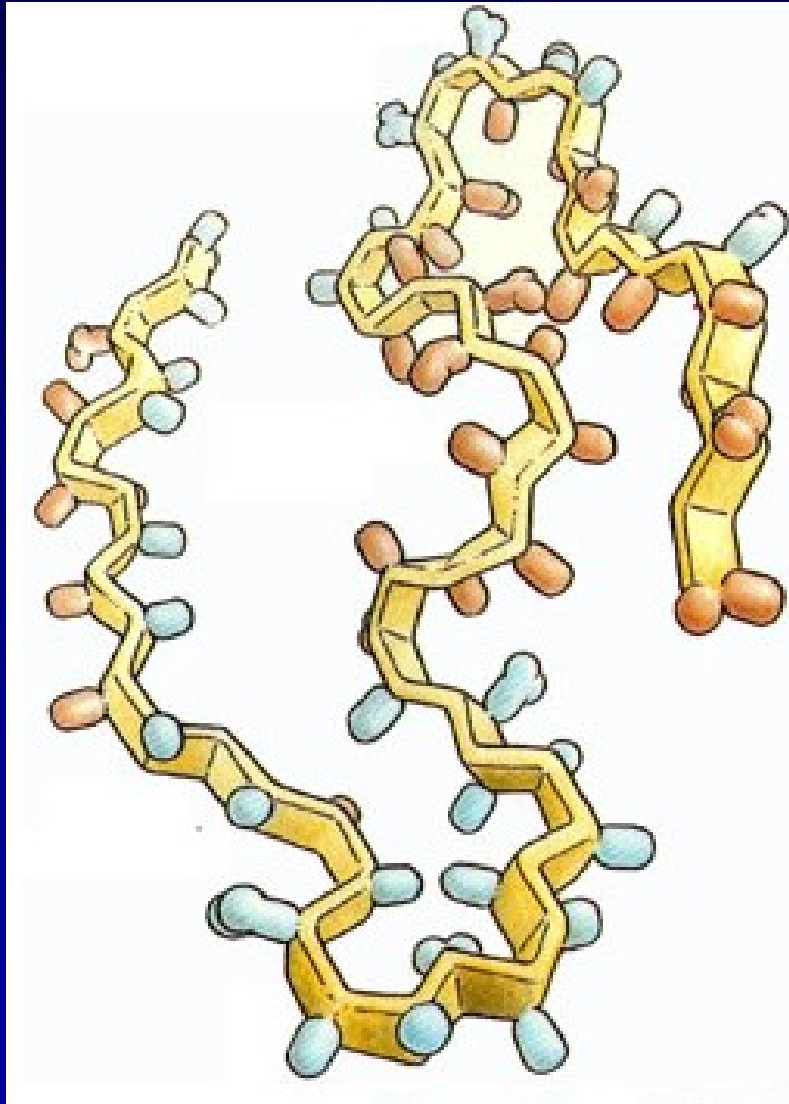
helix-coil transition of protein



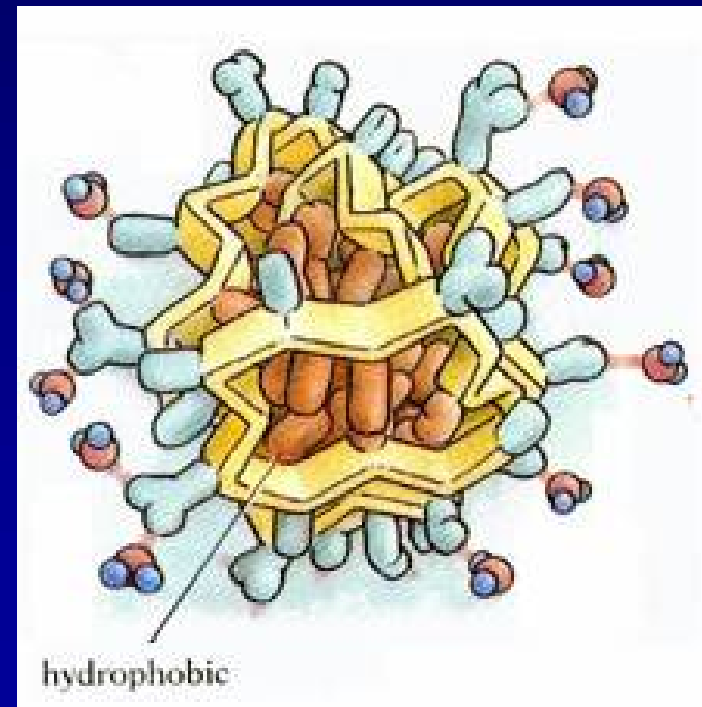
helix-coil transition of protein

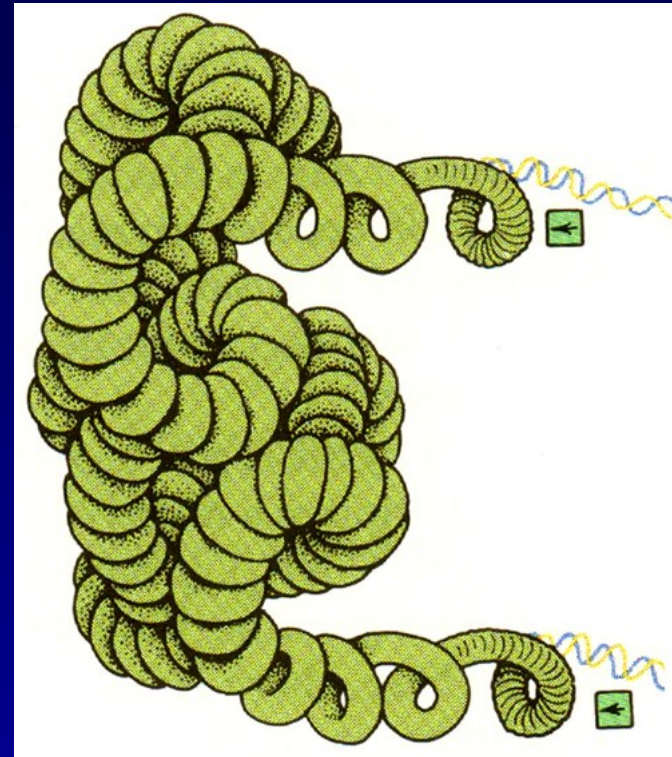
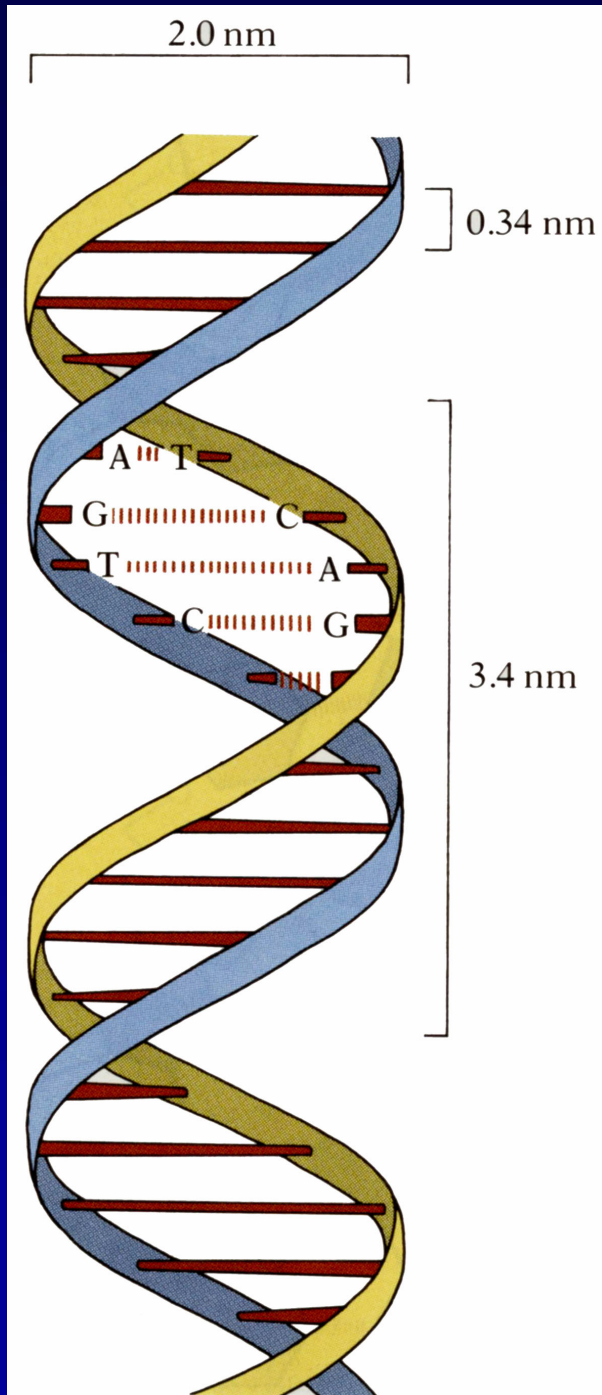


Folding of protein



hydrophilic 亲水的
hydrophobic 疏水的





highly coiled DNA

65~70°C

